Early Childhood Development and Structural Change

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Abstract

This paper presents a theory of how early childhood development shapes structural change. I document two novel cross-country facts: parental time devoted to young children (ages 0–4) and early childhood cognitive outcomes both rise with economic development, alongside the High-Skill Services employment share. I build a heterogeneous agent general equilibrium model that combines a multi-stage human capital technology with dynamic complementarity with non-homothetic preferences, unbalanced sectoral TFP growth and skill-biased technical change. Higher early childhood human capital produces a more skilled workforce, shaping structural change. The model matches time use, structural change and the skill distribution in the postwar US, and dynamic complementarity accounts for 29.8% of US structural change toward High-Skill Services. Finally, paid parental leave parameterized to Germany's *Elterngeld* accelerates structural change by 8.0%, despite general equilibrium effects muting it, suggesting that it can serve as a form of industrial policy.

JEL Codes: E24, J13, J22, J24, O11, O41

Keywords: early childhood development, structural change, high-skill services, time allocation, dynamic complementarity, skills

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1. Introduction

As countries develop, economic activity shifts away from Goods and into Services, a process known as structural change (Kuznets, 1957; Herrendorf, Rogerson and Valentinyi, 2014). Between 1975 and 2010, almost all countries in the GGDC 10-Sector Database experienced an increase in their Services employment share, and in roughly half of them the share rose by more than 20 percentage points (Timmer, de Vries and de Vries, 2015).[1] In developing countries, this shift has occurred at lower levels of income than in richer countries (Rodrik, 2016), making Services the central destination of labor for modern economies. The two leading explanations of structural change attribute this reallocation to non-homothetic preferences and differences in sectoral TFP growth rates.[2] Less attention has gone to understanding the role of the supply of human capital in shaping structural change, even though sectors differ in their skill intensity and so their productive capacity is constrained by the supply of skills. The papers that study this link model human capital either as a single schooling decision or a cohort-level endowment (Caselli and Coleman, 2001; Herrendorf and Schoellman, 2018; Porzio, Rossi and Santangelo, 2022). In doing so, they overlook a large body of work showing that skills are formed over multiple stages of childhood (Cunha et al., 2006; Cunha and Heckman, 2007; Cunha, Heckman and Schennach, 2010).

In this paper, I address this gap by asking how early childhood development shapes the sectoral composition of aggregate output. I pay special attention to the composition of Services, distinguishing between High-Skill and Low-Skill Services. This distinction matters because narrowing cross-country income differences will require that the transition to Services be accompanied by the reallocation of labor into skill-intensive, high-productivity industries that can sustain higher wages and better living standards (Duarte and Restuccia, 2010). To answer this question, I build on existing theories of structural change by endogenizing the supply of skills through early childhood development so that early childhood human capital, income and sectoral output are jointly determined in general equilibrium.

Central to my argument is the notion of dynamic complementarity: higher levels of human capital in *earlier* stages of childhood make investments in *later* stages of childhood more productive (Cunha and Heckman, 2007). Thus, the effects of early investments on childhood human capital not only persist over the life cycle, but they also amplify the

[1] These include advanced economies such as France (24.3 percentage points) and Italy (23.5), as well as developing economies such as Botswana (27.0), Brazil (26.2) and Thailand (22.8).

[2] The former argues that non-homothetic preferences induce households to shift expenditure toward luxuries as incomes rise (Kongsamut, Rebelo and Xie, 2001; Boppart, 2014; Comin, Lashkari and Mestieri, 2021; Alder, Boppart and Müller, 2022). The latter argues that unbalanced sectoral TFP growth raises the relative price of slow TFP-growing sectors and, given complementarities in sectoral demand, shifts activity towards them (Baumol, 1967; Ngai and Pissarides, 2007).

incentives for human capital accumulation in subsequent stages of childhood development. Higher early childhood human capital is therefore key to generating a more skilled workforce. A more skilled workforce shapes structural change by facilitating an increase in the productive capacity of the skill-intensive sector.

To motivate my theory, I document two novel cross-country empirical facts that coincide with the rise in the supply of skills (Katz and Murphy, 1992; Acemoglu and Autor, 2011) and the High-Skill Services employment share with development (Buera and Kaboski, 2012; Buera et al., 2022). First, using harmonized time use data (Lamote de Grignon Pérez et al., 2024), I show that conditional on living with a young child (ages 0–4) in the household, working-age individuals in richer countries tend to allocate more time to the childcare of young children.[3] I verify that this pattern is along the intensive margin (i.e. per child) and robust across mothers and fathers. Second, I use harmonized microdata (Bolgrien et al., 2024) to show that the cognitive outcomes of 4-year-olds improve with development.

I then embed my mechanism into a general equilibrium, dynastic overlapping generations model of growth and structural change with heterogeneous households and skill-biased technical change. The model serves two purposes: to quantify the contribution of early childhood development to structural change and to provide a laboratory for evaluating early childhood policies as a form of industrial policy. In the model, parents allocate their time between work and investing in their children's human capital, which evolves according to a multi-stage technology à la Del Boca, Flinn and Wiswall (2014). Time with children yields both a static utility benefit, which enters preferences as a luxury, and a dynamic benefit through the children's future human capital and ultimately their utility. Labor income is spent on output from three sectors: Goods, High-Skill Services and Low-Skill Services. The sectors differ in skill intensity and exogenous TFP growth. When children become adults, their human capital is fixed and, given wages, they choose whether to work as skilled or unskilled labor.

The supply-side mechanics of the model are driven by two forces and amplified by dynamic complementarity. First, as TFP rises and incomes grow, non-homothetic preferences over consumption and time induce parents to invest more time in their children — a key input to early childhood human capital. Second, as parental human capital rises across generations, so does the productivity of each hour parents invest in their children's human capital. Greater parental time investment improves early childhood human capital

[3]Parental time allocation is an important input for early childhood human capital. Using a structural model, Del Boca, Flinn and Wiswall (2014) find that parental time allocation from both parents is the most important input for the cognitive development of young children. The importance of parental time allocation falls with the age of the child at the expense of other inputs. Furthermore, Yum (2023) finds that both parental time allocation and other inputs are poor substitutes in early stages of childhood development. Other papers that stress the role of parental time allocation for early childhood human capital outcomes relative to other inputs include Francesconi and Heckman (2016), Del Bono et al. (2016), Bernal (2008) and Fiorini and Keane (2014).

and, through dynamic complementarity, generates a more skilled workforce. This raises the productive capacity of the High-Skill Services sector. Simultaneously, the traditional demand-side drivers of structural change are at play. The sectoral consumption aggregator is non-homothetic à la Comin, Lashkari and Mestieri (2021) such that as households get richer, parents bias their consumption towards luxuries (High-Skill Services) and away from necessities (Goods).

In general equilibrium, the distribution of human capital is key. On the one hand, the distribution of human capital shapes the household income distribution which, together with relative prices, drives aggregate *demand* for High-Skill Services. On the other hand, the human capital distribution shapes the supply of skills in the economy and, ultimately, the aggregate *supply* of High-Skill Services given its higher skill intensity. These forces, coupled with skill-biased technical change, jointly determine the return to skills. Together with parental preferences, this return shapes how much forward-looking parents invest in their children's human capital. Accordingly, the model captures the endogenous feedback effects between the supply- and demand-side drivers of structural change.

I then discipline the model to match patterns of structural change, time use and the skill distribution in the postwar United States (US). With the quantitative version of the model in hand, I perform several counterfactual exercises. One exercise asks whether differences in productivity can rationalize the cross-country empirical facts that motivate the model, namely the rise in parental time allocated to the childcare of young children, early childhood human capital and the High-Skill Services employment share with economic development. I implement this exercise by lowering initial TFP in each sector — the model's main lever of economic development — to match the US–South Africa and US–Zimbabwe TFP gaps in the 1960s (Feenstra, Inklaar and Timmer, 2015), holding all other parameters at their estimated values. I find that not only do differences in initial TFP rationalize cross-country differences in structural change, but they also compound over time; a 25% drop in initial TFP across the board reduces the human capital of 4-year-olds by 2.0% initially but by 9.4% after five generations (approximately seven decades). Similarly, the gap in the High-Skill Services employment share rises from 3.0 percentage points to 5.1 percentage points over the same time horizon. Accordingly, TFP gaps generate differences in early childhood human capital and structural change that compound over time.

A second exercise measures the role of dynamic complementarity across stages of childhood development in shaping US structural change. It quantifies how the cross-stage amplification embedded in the human capital technology affects structural change by shaping parents' incentives to invest in their children's human capital. To isolate this, I sever the link between stages of childhood: human capital at each stage is produced from the child's human capital draw at birth (genetics) rather than from the human capital

accumulated in the previous stage. Thus, investments at each stage no longer build on those made earlier. I find that once dynamic complementarity is no longer embedded in the technology, the High-Skill Services employment share in the US rises by 15.8 instead of 22.5 percentage points in the baseline. Put differently, dynamic complementarity explains 29.8% of the US's transition to the High-Skill Services sector. This measure accounts for the general equilibrium effects from the decline in the supply of skills and, consequently, the faster increase in the High-Skill Services sector's relative price which dampens the partial equilibrium effect of the decline in the supply of skills. The main takeaway from this exercise is that dynamic complementarity embedded in the human capital accumulation technology makes parental time investments in early childhood quantitatively important for structural change.

Finally, I use the quantitative version of the model as a laboratory for policy experiments. The quantitative exercises confirm that rising early childhood human capital has contributed non-negligibly to the sectoral reallocation of economic activity in favor of High-Skill Services in the postwar US. A natural extension is to consider what role, if any, policy can play in shaping structural change. That is, can childcare policies serve as a form of *industrial policy* that can influence the long-term composition of aggregate output? To proceed, I experiment with paid parental leave policies: a payroll tax-financed benefit that covers a fixed share of forgone earnings of parents who reduce their working hours to care for a newborn. The experiment requires altering the general equilibrium model by incorporating a government that collects taxes and administers the policy, which is parameterized to match the key features of Germany's *Elterngeld* program.

The benchmark policy experiment — which replaces two-thirds of forgone earnings of parents of newborns for up to one year — generates several conclusions. First, the paid leave policy accelerates the rise in the High-Skill Services employment share by 8.0% over five generations. This is small compared with the 25.6% faster rise in the physical quantity of High-Skill Services over the same horizon. This difference arises from general equilibrium effects that, because of the rise in the supply of skills, reduce the skill premium and the relative price of the High-Skill Services sector. Second, welfare as measured by consumption-equivalent variation rises by 9.9% by the sixth generation compared to a 2.5% loss experienced by the first generation that receives the leave benefit. Accordingly, the policy generates long-run gains at a short-run cost. Third, for a similar tax rate, a longer leave raises welfare by 15.2% compared with a higher replacement rate which raises welfare by 12.0%. The policy design is therefore key to maximizing the returns from the policy.

Related Literature. This paper contributes to two strands of literature. The first strand of literature is on the sources of structural change. In this paper, I focus on the role of

human capital in structural change, and therefore contribute to a growing literature on supply-side drivers of structural change (Caselli and Coleman, 2001; Porzio, Rossi and Santangelo, 2022; Herrendorf and Schoellman, 2018; Acemoglu and Guerrieri, 2008). In this respect, the closest paper to mine is Porzio, Rossi and Santangelo (2022). They model structural change in the context of an overlapping generations structure, whereby workers in different cohorts self-select into either agriculture or non-agriculture. Human capital in the model is drawn from a distribution that is a function of a cohort-specific component that evolves with the return to skills as well as an individual component that is exogenously determined. In contrast, I endogenize the human capital distribution through parental time investments, allowing my model to capture the two-way feedback between structural change and human capital. Furthermore, I enrich my framework with a multi-stage law of motion for human capital. That way, the model is well-equipped to study the effects of early childhood development on structural change.

The second strand of literature relates early childhood human capital to macroeconomic outcomes (Yum, 2023; Daruich, 2018; Lee and Seshadri, 2019; Manuelli and Seshadri, 2014; Schoellman, 2016; Caucutt and Lochner, 2020; Youderian, 2019; de la Croix and Licandro, 2013). This literature is motivated by growing evidence that early childhood development is influential in driving life-cycle outcomes (Deming, 2009; García et al., 2017; Cunha and Heckman, 2007; Cunha, Heckman and Schennach, 2010). Of these papers, the one closest to mine is Schoellman (2016), which uses variation in the ages at which Indochinese migrants arrived in the US to study the effects of an additional year of early childhood in a rich economy setting. The findings suggest that early childhood development is not a strong determinant of cross-country income differences. My paper is complementary in the sense that rather than the country environment, I focus explicitly on the importance of parental time inputs for early childhood human capital and, ultimately, structural change. This is justified by the growing body of evidence that stresses the importance of parental childcare time for early childhood human capital outcomes (Francesconi and Heckman, 2016; Del Boca, Flinn and Wiswall, 2014; Del Bono et al., 2016; Guryan, Hurst and Kearney, 2008; Fiorini and Keane, 2014; Bernal, 2008).[4]

[4]Parental time allocation is known to shape the human capital formation of children, especially in early stages of childhood. This link operates through several channels. One such channel is through language acquisition and cognitive stimulation, as evidenced by Hart and Risley (1995) and Villena-Roldan and Ríos-Aguilar (2012). Another such channel is through problem-solving abilities; through prompts and structure, parents are able to calibrate support in real time to their children such that they can solve problems beyond their unassisted capacity. See Vygotsky (1978) for a theoretical discussion and Pratt et al. (1988) and Neitzel and Stright (2003) for empirical evidence. A final channel operates through stress regulation. Blair and Raver (2012) and Belsky and Fearon (2002) both discuss the role of sustained sensitive caregiving and attachment security for the development of executive functions and cognitive competence of children.

Outline. The rest of the paper is organized as follows. Section 2 presents the empirical facts that motivate the theory. Section 3 presents the full general equilibrium model with heterogeneous agents, multiple sectors, skill-biased technical change and endogenous human capital accumulation. Section 4 calibrates the model while Section 5 uses the quantitative version of the model to perform the counterfactual exercises. Section 6 explores the implications of paid parental leave as a form of industrial policy. Section 7 offers concluding remarks.

2. Empirics

This section documents novel cross-country evidence on parental time investments in young children (ages 0–4) and the cognitive outcomes of 4-year-old children to motivate the general equilibrium model in Section 3.

2.1 Data Sources

This section compiles data from four sources. The first data source is the latest version (v11) of the Multinational Time Use Survey (MTUS) (Lamote de Grignon Pérez et al., 2024). The MTUS is a harmonized time use survey that contains records on how diarists spent 1440 minutes on a given single day. The MTUS has two types of data. The first is the harmonized aggregated file which contains day-level data (i.e. how much diarists spent on each activity in a single day) while the second is the harmonized episode file data which contains episode-level data (i.e. an episode is a change in activity, location, transport, co-presence or the use of technological devices). I rely on the harmonized aggregated file given the interest in aggregate time use trends. The time use data is accompanied by a number of useful variables including the age, sex and education level of the diarist as well as data on the number of household members, the number of children below the age of 18 living the household and the age of the youngest child living in the household. Accordingly, each observation in the harmonized aggregated file data corresponds to a country-year-person-day observation.

The second data source is the United Nations International Children's Emergency Fund's (UNICEF) Multiple Indicator Cluster Survey program accessed through the Integrated Public Use Microdata Series (IPUMS-MICS) (Bolgrien et al., 2024). IPUMS-MICS offers harmonized cross-country survey-level data on children worldwide. The dataset focuses on a diverse set of topics, including early childhood development. The IPUMS-MICS dataset is split into multiple units of analysis including births, children between the ages of 0 and 4, children between the ages of 5 and 17 and women. The early childhood development

module contains data on parental time investments in children and early childhood human capital outcomes.

The third dataset is the IPUMS International (IPUMS-I) dataset (Ruggles et al., 2024). This dataset contains rich household- and person-level data that is harmonized across country-year samples. The list of variables recorded is large, ranging from nativity and birthplace to occupation and industry of work. I use the person-level census data with an explicit focus on the occupation and industry modules for information on the individual's primary occupation and industry of work.

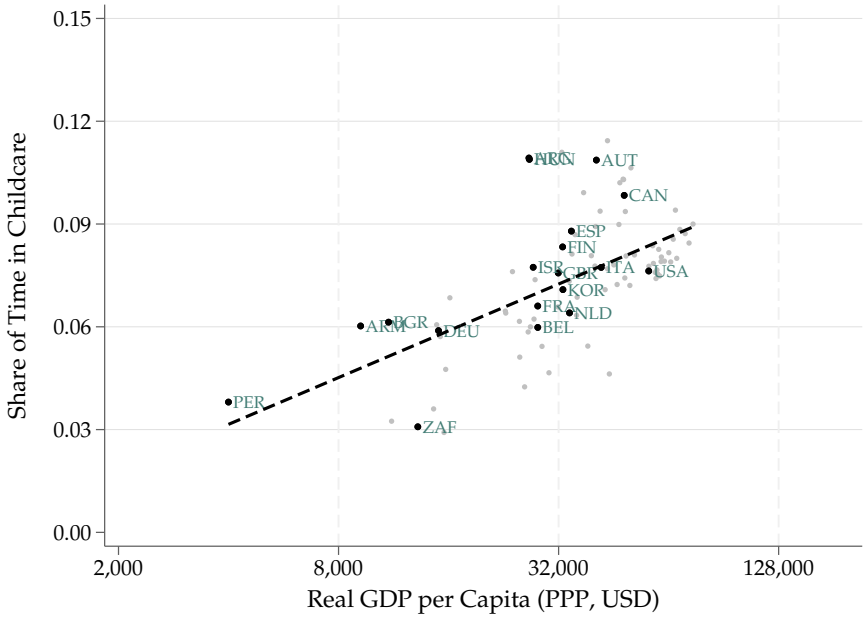
The fourth dataset is the Penn World Tables v11.0 (Feenstra, Inklaar and Timmer, 2015) which I use to obtain data on PPP-adjusted real GDP per capita at the country-year level — a standard proxy for development.

2.2 Cross-Country Patterns

To motivate my theory, I begin by documenting the first novel fact: among individuals living with a young child (ages 0–4), time spent on childcare rises with economic development. Childcare in the time use survey is defined as physical and/or medical childcare, teach and/or help with homework, read, talk and/or play with child, and supervise, accompany and/or other childcare. I restrict my attention to individuals who are between 18 and 64 years of age (i.e. working-age individuals).[5] Figure 1 plots the country-year population-weighted average time spent on the childcare of young children by working-age individuals conditional on living with a young child against PPP-adjusted real GDP per capita. It shows that individuals in richer countries, on average, devote more time to the childcare of young children. To gauge the magnitude of these cross-country income differences, Americans earned four times as much as South Africans on average in 2010, and spent two and a half times as much time on childcare in the same year (almost two hours compared to forty two minutes).

[5]It is not possible to identify the diarist's relationship with the child (i.e. whether the diarist is the mother, father, sibling, etc. of the child). The only observables are the age of the youngest child living in the household and the number of children (below 19 years of age) in the household.

Figure 1. Share of Time Spent on Childcare



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

Figure 1 differs from other attempts to document cross-country patterns of parental time investment in childcare. Relative to Bick, Fuchs-Schündeln and Lagakos (2018) and Guryan, Hurst and Kearney (2008), I focus on early childhood human capital inputs. This means that I restrict my attention to diarists who live with at least one young child (ages 0–4) in the household rather than children under 18.

This fact can benefit from a number of robustness checks. To ensure that this trend is driven by per-child time investments, I also plot the same variable with an additional restriction that the number of children in the household is 1. Appendix A.1 shows that, qualitatively, the same trend holds, ruling out any conclusion relating this novel fact with demographic trends. Another potential challenge to Figure 1 is that while offering an expanded set of countries relative to previous attempts, it is not clear if this fact is robust across the lower end of the development spectrum. Additionally, one limitation of time use data is that it does not clarify the nature of the relationship between the individual and the child. To tackle both challenges, I present data from the IPUMS-MICS. This dataset asks mothers and fathers of children aged between 0 and 4 years of age six binary questions each on whether or not the parent in question has undertaken a given time-intensive activity with that child in the last 3 days. After aggregating the binary responses for both parents at the country-year level using children under 5 sample weights, I plot these against

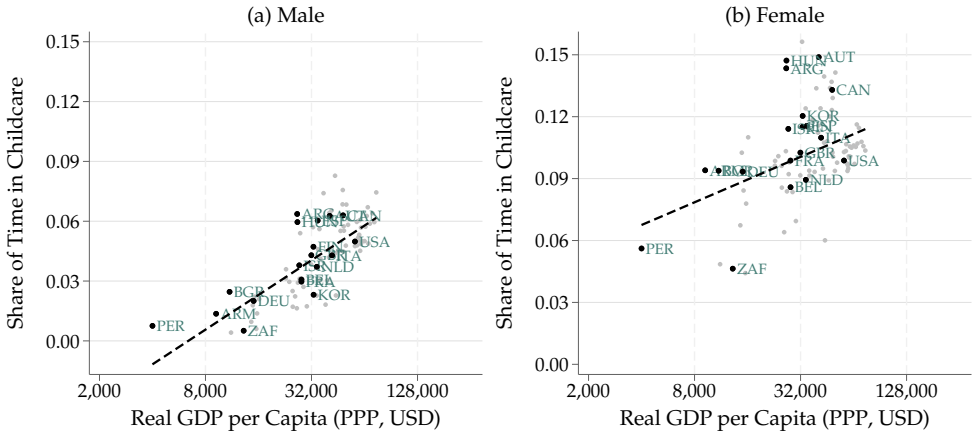
PPP-adjusted real GDP per capita.[6] The results are presented in Appendix A.2 and confirm the empirical fact that parents in richer countries dedicate more time to early childhood childcare, even on the lower end of the development spectrum. It is also helpful to confirm that these results hold within-country. While it is a well-established fact in the literature that time allocated to childcare rises over time in the US (Aguiar and Hurst, 2007; Guryan, Hurst and Kearney, 2008; Ramey and Ramey, 2010), I present the plot in Appendix A.3 using the same data as Figure 1 (focusing on children between the ages of 0 and 4 inclusive) for the sake of completeness. Barring 1975, there is a consistent increase in time spent on childcare over time implying that this phenomenon is observed both within and across countries over the development process.

Figure 2 plots the share of time spent on childcare against PPP-adjusted real GDP per capita by gender. It shows that on average both men and women dedicate more time to the childcare of young children as the economy develops. Appendix A.4 verifies that these gender-specific trends also arise from the IPUMS-MICS dataset. The difference in levels between men and women is not surprising given that these trends coincide with a number of important gender-biased structural changes accompanying the development process including changes in female labor force participation (Goldin, 1994) and home production (Bridgman, Duernecker and Herrendorf, 2018). However, that the time allocated to the childcare of young children rises between females suggests that this pattern could be separated from female labor force participation trends given that greater participation should intuitively induce a decline in time allocated to the childcare of young children. Appendix A.5 shows that the childcare time allocation share of females increase with development regardless of employment status, suggesting that this fact is robust to female labor force participation trends.

I then document a second novel fact: early childhood cognitive skills improve with development. To the best of my knowledge, this is the first attempt to do so at the cross-country level. To measure these outcomes, I use the IPUMS-MICS dataset which asks the child's primary caretaker three binary questions: (i) can the child read at least four simple, popular words? (ii) can the child identify at least 10 letters of the alphabet? and (iii) can the child name and recognize symbols of all numbers from 1-10? Aggregating the number of yes values at the child-level (assigning equal weights to each question) and then aggregating again at the country-year level using the proposed under-5 survey weights yields a value from 0 to 3 for each country-year observation. I plot these against PPP-adjusted real GDP per capita in Figure 3. Since the speed of cognitive development

[6]The IPUMS-MICS dataset does not offer any data on minutes spent on childcare by parents, but the literature on early childhood development notes that parental time investments can be (and are frequently) measured by the frequency of activities (see Attanasio, Cattan and Meghir, 2022).

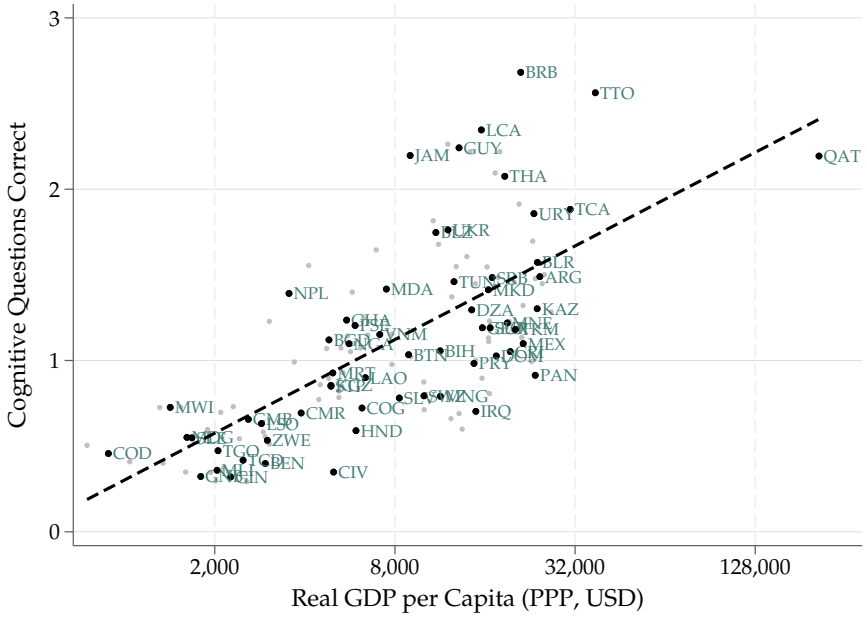
Figure 2. Share of Time Spent on Childcare By Gender



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

can be closely related to age when kids are youngest, I separate children that are 3 years of age and children that are 4 years of age.

Figure 3. Early Childhood Cognitive Development Demonstrated



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

The correlation between both variables is visibly strong. In Thailand, the average PPP-adjusted real GDP per capita is roughly eight times the size of the real GDP per capita

in Chad in 2019. The average Thai four-year-old answers two of the three questions correctly, while their Chadian counterpart is more likely than not to answer none. This example not only highlights that cross-country differences exist in early childhood development outcomes, but that these differences are sizable even when comparing two countries at the lower end of the development spectrum. Since the development of cognitive skills is sensitive to age when children are young, Appendix A.6 plots the same chart for three-year-old children. The results are qualitatively similar.

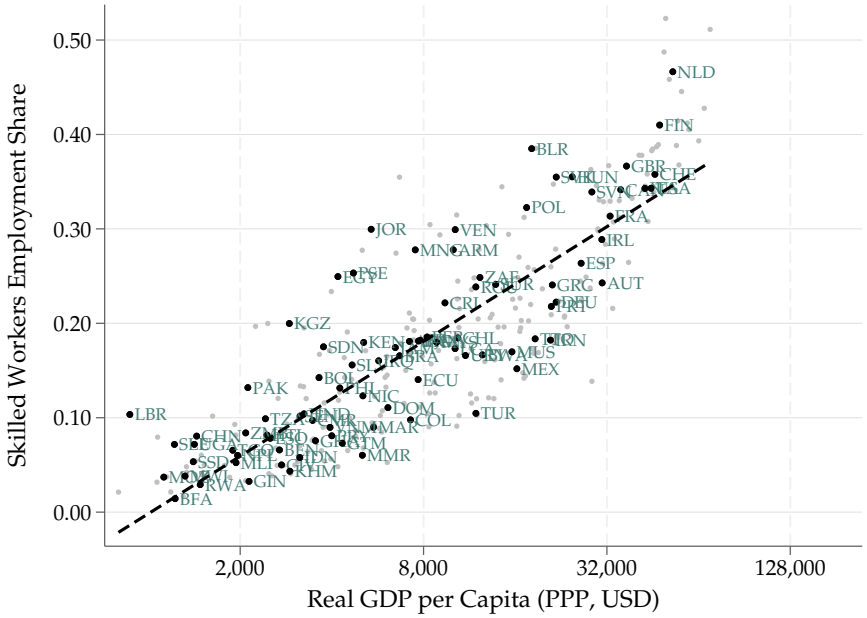
Thus far, I have shown that both time allocated to the childcare of young children and early childhood cognitive skills are correlated with development. To motivate the link between early childhood development and structural change, I use IPUMS-I (Ruggles et al., 2024) to show that both facts coincide with the rise in the supply of skills and the High-Skill Services employment share with development. Figure 4 plots the population-weighted share of total employment of skilled workers against PPP-adjusted real GDP per capita. Skilled workers are defined, following Acemoglu and Autor (2011), as workers who belong to managerial, professional or technical occupations.[7] In contrast to educational attainment — a common proxy for a worker's skill level — the occupation captures the multi-dimensional nature of skill content, including educational attainment and wage levels. Figure 4 shows that along the development path, the share of skilled workers rises. This fact has been documented before; Katz and Murphy (1992), Acemoglu and Autor (2011) and Buera et al. (2022) are among the papers that have done so in part to understand why the wage premium rises with the relative supply of skilled labor. Figure 4 is instead helpful in showing that this phenomenon holds for a broad set of countries.

I then split Services industries in the INDGEN variable into High- and Low-Skill Services sectors based on the industry's share of total employment coming from skilled occupations, focusing only on the US censuses as the benchmark. Table A.1 in the Appendix presents the results of the industry classification by the 1960-2020 average skilled share of employment. There is a clear gap in the average skilled share of employment between the first four and the latter six Services industries, generating a natural threshold that places Education, Health and Social Work, Financial Services and Insurance and Business Services and Real Estate in the High-Skill Services bin. This classification appears consistent with every individual census year in addition to the 1960-2020 average.

Figure 5 confirms the unambiguous positive correlation between the High-Skill Services share of employment and development documented by Buera and Kaboski (2012) and Buera et al. (2022). This is not surprising given the current state of knowledge on the process of structural transformation; the early works of Kuznets (1957), Fuchs (1968) and

[7]These are identified as observations whose primary occupation is either legislator/senior official/manager, professional or technician/associate professional which is identified by the OCCISCO variable.

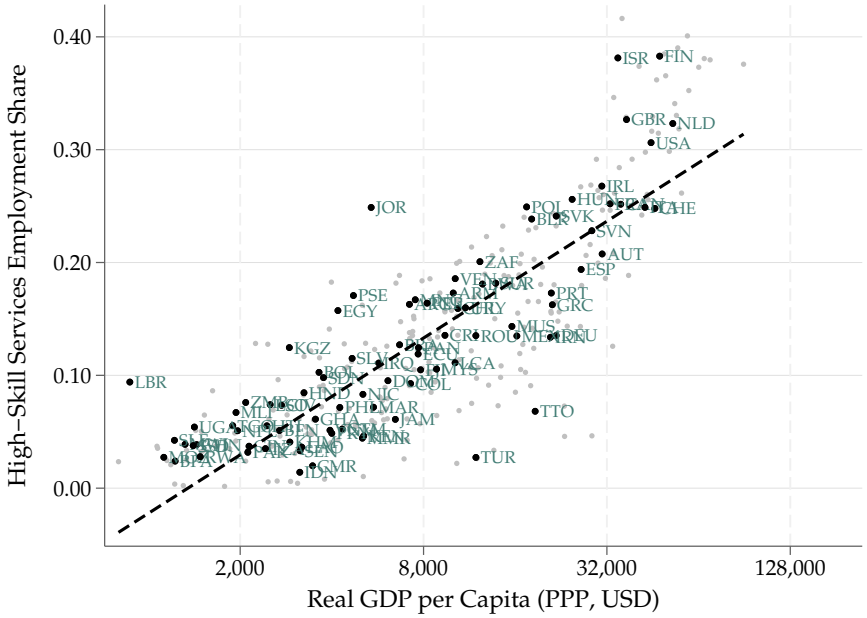
Figure 4. Skilled Workers Employment Share



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

Baumol (1967) among others showed a shift in economic activity towards Services over the development path. More recent papers such as Buera and Kaboski (2012), Fang and Herrendorf (2021) and Buera et al. (2022) have focused exclusively on the sources and consequences of the growing importance of High-Skill Services with development, and so this is not the first paper to document the growing importance of High-Skill Services accompanying the process of growth and development. Like Figure 4 however, Figure 5 is helpful for showing that this phenomenon holds for a broad panel of countries across the development spectrum.

Figure 5. High-Skill Services Employment Share



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

2.3 Summary

This section then documents two novel empirical facts on early childhood human capital inputs and outputs. The two novel facts are that parents in richer countries on average dedicate more time to their young children and these young children perform better in cognitive tasks. This section then motivates the interest in High-Skill Services by showing that both the supply of skills and the High-Skill Services employment share rise with development, with both eventually taking up two-fifths of total employment in the richest countries.

3. Model

To uncover and quantify the causal link between early childhood development and structural change, I construct a heterogeneous agents general equilibrium framework. The exercise takes a standard multi-sector model of growth and structural change with skill-biased technical change in the spirit of Buera et al. (2022) and endogenize the distribution of skills using a multi-stage human capital accumulation technology with dynamic complementarity à la Del Boca, Flinn and Wiswall (2014). The model jointly accounts for time use, early childhood human capital, and structural change patterns observed in the data.

3.1 Environment

Assume an infinite time horizon where time is denoted by $t \in \mathbb{Z}_{\geq 0}$. At any given time t, there are three sectors that operate in this economy: Goods (G), Low-Skill Services (L) and High-Skill Services (H). Each sector has a representative firm that operates in a perfectly competitive environment, and each firm decides how many efficiency units of skilled (S) and unskilled (N) labor to hire in their respective static profit-maximization problems.

Let the economy be populated by a fixed continuum of dynasties i of measure $\mathcal{L}$ and let $g \in \{1, ..., \infty\}$ serve as a generation superscript. At any given time t, a dynasty consists of a single parent and a single child (also referred to as a household hereafter).[8] The parent is the only decision-maker inside the household. Children do not work nor consume, parents invest time to develop their child's human capital because they derive direct utility (i.e. enjoyment) from it and because they attach value to their child's future outcomes. Parental time investments are the only human capital investments allowed in this economy, and the elasticity of this input varies with the child's stage of human capital development.

Time allocated to childcare has an opportunity cost of time at work, which earns the income that parents spend on output from three sectors.[9] A parent working as a skilled worker contributes their human capital level as efficiency units of labor per unit of time while all parents supply a single efficiency unit of labor per unit of time if they work as an unskilled worker regardless of their human capital level and the sector they are employed in. At any given time t, parents belong strictly to one sector and occupation. Accordingly, at all times the incumbent parent of household i seeks to maximize utility by solving the following three problems: (i) a decision on how much time to invest in their child; (ii) a decision on how much to consume from each sector; and (iii) a decision on which occupation $\{S, N\}$ and sector $\{G, L, H\}$ to work in.

All households live hand-to-mouth in this economy as there are no financial markets. There is free labor mobility across both sectors and occupations for workers, and all markets clear at all times.

Timing and Heterogeneity. Each individual lives for $2T$ years. For the first T years, the individual is a child. For the subsequent T years, the individual is a parent to a child of their own.[10] The cycle repeats perpetually for each dynasty over the infinite time horizon. At any time t, an individual belonging to a given generation g is $t - \left[(g-1) \times T\right]$ years old. Let

[8] This choice of demographics is motivated by the empirical facts; the rise in the share of time parents spend with their children occurs on the intensive margin (i.e. per child) and is observed for both mothers and fathers.

[9] I therefore use the terms "worker" and "parent" interchangeably.

[10] The life-cycle notation and structure here is similar to (Cunha and Heckman, 2007).

$\mu(h, h'; t)$ denote the joint probability density function at time t of a generic parent-child human capital pair $h_i^g = h$ and $h_i^{g+1}(t) = h'$.

Upon birth, the child of dynasty i draws an initial human capital level $h_i^{g+1}(t = gT)$ from a lognormal distribution $G(\eta_c, \kappa_c)$ where η_c controls the location and κ_c controls the scale of the distribution. This distribution is time- and dynasty-invariant, and is known to all parents. Furthermore, each parent observes their child's draw upon birth. The human capital of the child from birth to age T evolves through the multi-stage law of motion for human capital accumulation, after which time the human capital level is fixed for the remainder of the individual's life. Consequently, at any time t, households are ex-ante heterogeneous in parent-child human capital pairs $h_i^g, h_i^{g+1}(t) \in \mathcal{H} \times \mathcal{H}$.

Households. At every time t, each parent with parent-child human capital pair $h_i^g, h_i^{g+1}(gT)$ is endowed with a single unit of time to be allocated between two activities: working $n_i^g(t)$ and spending time with the child $\tau_i^g(t)$. Parents seek to maximize the present discounted utility at all times. For a given dynasty, I specify the following discrete-time life-cycle structure in recursive formulation for preferences from the point of view of a parent of household i belonging to generation g

$$V_i^g\left(h_i^g, h_i^{g+1}(gT)\right) = \max_{\{\tau_i^g(t)\}_{t=gT}^{(g+1)T-1}} \Bigg\{ \underbrace{\sum_{t=gT}^{(g+1)T-1} \beta^{t-gT} U\left(c_i^g(t), \tau_i^g(t)\right)}_{\text{current generation utility}} + \beta^T \beta_g \underbrace{\mathbb{E}_{h_i^{g+2}}\left[V_i^{g+1}\left(h_i^{g+1}((g+1)T), h_i^{g+2}((g+1)T)\right)\right]}_{\text{future generations' utility}} \Bigg\}, \tag{1}$$

where β is the time discount factor, β_g is the inter-generational discount factor and $c_i^g(t)$ is the household's aggregate consumption. It is easy to see that the current formulation for preferences suggests that the parent internalizes the utility of future generations of the dynasty: the first term is the present discounted value of utility of the parent belonging to generation g while the second term is the continuation value which collects the sum of lifetime utility of *all* descendants discounted by an additional inter-generational discount factor β_g. The parent of generation g only observes the human capital draw of their child of generation $g+1$ and not that of the generation $g+2$ child. Accordingly, the parent takes expectations over the latter using the time-invariant initial human capital distribution $G(\eta_c, \kappa_c)$ known to the current parent.

The utility function is concave, strictly increasing and twice differentiable with $U : \mathbb{R}^+ \rightarrow \mathbb{R}^+$. I assume the familiar MaCurdy (1981) structure for instantaneous utility[11]

$$U\left(c_i^g(t), \tau_i^g(t)\right) = \frac{c_i^g(t)^{1-\nu}}{1-\nu} - \omega \frac{n_i^g(t)^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}}. \tag{2}$$

The parameter ω is the time-invariant weight that the parent attaches to the pure dis-satisfaction generated from work. The choice to model a disutility for hours worked (i.e. utility from time spent with children, akin to leisure) in the first place is motivated by the findings of Cosaert and Hennebel (2023) who, using Dutch consumption and time use data, find that 40% of parental childcare time allocation is perceived as a form of leisure. ν is the usual CRRA parameter which controls the curvature in consumption. It can be shown that this parameter will determine the strength of income effects; if $\nu > 1$, then a rise in a given parent's income will prompt a decline in hours worked and vice versa.

Aside from their time allocation, the parent also chooses the sectoral composition of aggregate consumption. Letting $c_i^g(t)$ be a composite of consumption from three sectors: Goods ($c_{i,G}^g(t)$), Low-Skill Services ($c_{i,L}^g(t)$) and High-Skill Services ($c_{i,H}^g(t)$), I assume the following implicitly defined CES aggregator from Comin, Lashkari and Mestieri (2021)

$$\sum_{j \in \{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}} \right)^{\frac{\sigma-1}{\sigma}} = 1, \tag{3}$$

where θ_j is a time-invariant sector-specific preference weight, $c_{i,j}^g(t)$ is the sector-specific quantity consumed and $\sigma \in (0, \infty)$ is the (constant) elasticity of substitution that governs the degree of substitutability (if $\sigma > 1$) or complementarity (if $\sigma < 1$) between sectoral consumption. Crucially, λ_j influences the relative income elasticities of sectoral demand. Consequently, this will govern the extent to which the sectoral composition of the consumption basket will vary with household income. The advantage of (3) relative to alternative consumption aggregators such as Stone-Geary preferences is that, as will become clearer when studying the equilibrium path of this economy, income effects do not level off but rather persist in the long-term, better capturing long-run Engel curves.

Finally, each parent self-selects into sector $\{G, L, H\}$ and occupation of work $\{S, N\}$ at any given time t. Parents who choose a skilled occupation contribute h_i^g units of efficiency labor while those that choose an unskilled occupation contribute exactly 1 unit of efficiency

[11]This utility function choice is consistent with the wider literature on time allocation and aggregate labor supply over time (see Bick et al. (2022) and Boppart, Krusell and Olsson (2023) for some examples of this). Boppart and Krusell (2020) show that MaCurdy (1981) preferences can generate a balanced growth path with falling hours worked with income.

labor regardless of their human capital level. Accordingly, the parent's gross income at time t, $y_i^g(t)$, belongs in the following set

$$y_i^g(t) \in \left\{ w_{j,S}(t) h_i^g, w_{j,N}(t) : j \in \{G, L, H\} \right\}, \tag{4}$$

where $w_{j,S}(t)$ and $w_{j,N}(t)$ are the wage for a single unit of efficiency labor as a skilled and unskilled worker in sector j, respectively. Since households live hand-to-mouth in this economy, the following budget constraint must hold at all times

$$\left(1 - \tau_i^g(t)\right) y_i^g(t) \geq p_i^g(t) c_i^g(t), \tag{5}$$

where $p_i^g(t)$ is the price of aggregate consumption.

Human Capital Accumulation. Inspired by Del Boca, Flinn and Wiswall (2014) and Cunha, Heckman and Schennach (2010), parents have access to a multi-stage human capital production function for their children. The child's initial human capital is drawn from a time- and dynasty-invariant lognormal distribution $G(\eta_c, \kappa_c)$ where η_c and κ_c control the location and scale of the distribution, respectively. Each parent knows the distribution and observes the child's draw at all times.[12] Let $s \in \{1, ..., S\}$ denote the stage of childhood development that a child is currently in. The child's human capital evolves according to

$$h_i^{g+1}(t+1) - h_i^{g+1}(t) = h_i^{g+1}(t)^{\zeta_s} \left(h_i^g \tau_i^g(t) \right)^{1-\zeta_s}, \tag{6}$$

where $t \in \{gT, \ldots, (g+1)T - 1\}$ indexes the periods of childhood and $s = t - gT + 1 \in \{1, \ldots, S\}$ denotes the corresponding stage of development. The first input $h_i^g \tau_i^g(t)$ is the parent's effective time allocation to childcare. This is composed of the units of time that the parent of household i and generation g chooses to spend with their child, $\tau_i^g(t)$, and the parent's fixed human capital stock h_i^g which acts as a productivity term to the childcare time allocation. The second input is the current human capital level of the child, $h_i^{g+1}(t)$. These inputs typically feature in multi-stage human capital models in a Cobb-Douglas fashion, as in Del Boca, Flinn and Wiswall (2014). Crucially, the relative intensities of these inputs (governed by $1 > \zeta_s > 0$) vary with the stage of childhood development, s, to capture the varying levels of importance associated with each input over the childhood development period. Another important feature of this technology is that it embeds the notion of dynamic complementarity: children with higher early childhood human capital

[12] An alternative modeling choice would be to assume that each parent does not observe the child's draw and instead makes decisions based on the expectation of the distribution such that the child's human capital is realized ex-post all investment decisions (i.e. when they become parents themselves). To remain consistent with the literature on childhood development (e.g. Del Boca, Flinn and Wiswall, 2014), I assume that the parent observes the child's draw.

draw in greater investments by their parents in subsequent stages of their childhood, fixing the parent's human capital level.[13] As soon as the child becomes a parent at the age of T, the human capital level is fixed for the remainder of life. For simplicity, there is no depreciation in human capital in this economy.

Production. On the production side[14], there is a profit-maximizing sector-specific representative firm that operates in a perfectly competitive environment. The firm produces sectoral output $Y_j(t)$ by augmenting aggregate efficiency units of labor $Z_j(t)$ using a sector-specific TFP $A_j(t)$ which grows at a constant and exogenous (sector-specific) growth rate g_j known to all parents at all times

$$Y_j(t) = A_j(t)Z_j(t) \quad \forall j \in \{G, L, H\}. \tag{7}$$

The aggregator $Z_j(t)$ represents a CES technology that combines both skilled, $S_j(t)$, and unskilled, $N_j(t)$, efficiency units of labor in sector j

$$Z_j(t) = \left[\phi_j N_j(t)^{\frac{\epsilon-1}{\epsilon}} + \left(1-\phi_j\right)\left(B(t)S_j(t)\right)^{\frac{\epsilon-1}{\epsilon}}\right]^{\frac{\epsilon}{\epsilon-1}}. \tag{8}$$

The skill intensity in sector j is controlled by ϕ_j. This is the second dimension along which sectors can differ, together with TFP. The elasticity of substitution between both types of labor is common across sectors and is governed by $\epsilon \in (0, \infty)$. If $\epsilon > 1$ ($\epsilon < 1$), then both types of labor are gross substitutes (complements) and the aggregator collapses to a Cobb-Douglas if $\epsilon = 1$. Skill-biased technical change is incorporated through $B(t)$ and grows at a constant rate g_B. This specification captures the rise in the productivity of skilled labor *relative* to unskilled labor over time. This is sector-invariant, meaning that it benefits skilled workers across all sectors equally. Output from sector $j \in \{G, H, L\}$ is sold to households at a price $p_j(t)$.

3.2 Equilibrium

Sectoral Producers. Each sectoral producer chooses the quantities of skilled, $S_j(t)$, and unskilled labor, $N_j(t)$, to maximize profits given their technologies, output prices and wages. Taking the ratio of first order conditions from profit-maximization implies that the relative demand for skilled labor in sector j can be written as

[13] Mathematically, dynamic complementarity can be verified by checking if the cross-partial derivative of the human capital production function $\frac{\partial^2 h_i^{g+1}(t+1)}{\partial h_i^{g+1}(t)\partial \tau_i^g(t)} > 0$.

[14] Since only the parent can work in this economy, the generation indices are dropped hereafter.

$$\frac{S_j(t)}{N_j(t)} = \left(\frac{1-\phi_j}{\phi_j}\frac{w_{j,N}(t)}{w_{j,S}(t)}\right)^{\epsilon} B(t)^{\epsilon-1}. \tag{9}$$

A few observations are in order. First, a higher relative wage rate of unskilled efficiency units of labor in sector j, $w_{j,N}(t)$, compared to skilled efficiency units of labor, $w_{j,S}(t)$, increases the relative demand of skilled labor relative to unskilled labor. The second observation is that a higher skill intensity $(1-\phi_j)$ increases the relative demand of skilled labor in sector j. The third observation is that the direction in which skill-biased technical change affects the relative demand of skilled work depends on the elasticity of substitution between both types of labor, ϵ. That is, assuming that skill-biased technical change is in favor of skilled labor (i.e. $g_B > 0$), then this increases (decreases) the relative demand of skilled labor only if both types of labor are gross substitutes (complements).[15]

Parents. Parents decide (i) which occupation and sector to work in, (ii) how much time to invest in their child and (iii) how much of each sector to consume given preferences, wages and prices. Proposition 1 simplifies the household's problem by proving that a utility-maximizing parent is also a gross income-maximizing parent.

Proposition 1 (Separation of occupational choice)**.** *Suppose that the occupational choice problem only affects the parent's problem through gross income. Then, the parent's problem can be solved sequentially. First, the parent makes the occupational choice that maximizes gross income. Second, the parent jointly chooses time allocation and sectoral quantities that maximize preferences (1) subject to (5) and* $n_i^g(t) + \tau_i^g(t) \leq 1$.

Proof. Any feasible allocation of time and sectoral quantity under gross income $y(t)$ remains feasible under a gross income $y'(t) > y(t)$ since $y'(t)$ expands the parent's budget without affecting preferences or the time constraint. See Appendix B.2 for formal proof. □

Proposition 1 allows (4) to be written as a standard Roy selection-type problem

$$y_i^g(t) = \max\left\{w_{j,S}(t)h_i^g, w_{j,N}(t) : j \in \{G, L, H\}\right\}, \tag{10}$$

where given wages and the fixed human capital level, parents choose the sector and occupation that maximizes their gross income. With gross income determined, the parent's problem boils down to a standard two-part structural change model where $\forall t$ the given parent chooses $\tau_i^g(t)$ to maximize intertemporal preferences (1) subject to the budget constraint (5) (i.e. the intertemporal problem) and then chooses the sectoral consumption quantities that maximize (3) subject to the budget constraint (5) with net income substituted

[15]The proof for (9) is located in Appendix B.1.

in (i.e. the static problem). The solution to the static problem yields the standard Marshallian demand functions[16]

$$c_{i,j}^{g}(t) = \frac{e_{i,j}^{g}(t) p_i^{g}(t) c_i^{g}(t)}{p_j(t)}, \quad \text{where} \quad e_{i,j}^{g}(t) = \theta_j \left(\frac{p_j(t)}{p_i^{g}(t)} \right)^{1-\sigma} c_i^{g}(t)^{(\sigma-1)(1-\lambda_j)} \quad \forall j \in \{G, L, H\}. \tag{11}$$

The demand functions write the sectoral consumption quantity from a given sector as a function of the parent's total expenditure, $p_i^{g}(t) c_i^{g}(t)$ (which by virtue of the budget constraint is equivalent to time-adjusted income $(1 - \tau_i^{g}(t) y_i^{g}(t))$. Note that the standard income and substitution effects are present in (11). A higher relative price $p_j(t)$ reduces consumption of sector j's good, while greater aggregate consumption induces a disproportionate change in consumption of sector j's good by a magnitude controlled by λ_j.[17]

Equilibrium. I set the price of the H sector's output $p_H(t)$ as the numéraire in this economy.

Definition 1 (Competitive equilibrium)**.** Normalize $p_H(t) = 1$ for all $t \in \mathbb{Z}_{\geq 0}$. A competitive equilibrium is a collection of price, wage, and technology sequences

$$\left\{ \left\{ p_k(t) \right\}_{k \in \{G,L\}}, \left\{ w_{j,o}(t) \right\}_{j \in \{G,L,H\},\, o \in \{S,N\}}, \left\{ A_j(t) \right\}_{j \in \{G,L,H\}}, B(t) \right\}_{t \in \mathbb{Z}_{\geq 0}},$$

together with household allocations and firms' labor demands, such that:

1. Given prices and wages, the parent in each dynasty i and generation g, with state $\left(h_i^{g}, h_i^{g+1}(gT) \right)$, maximizes intertemporal utility by choosing sequences of occupations, sectors of employment, sectoral consumption, and childcare time,

$$\left\{ o_i^{g}(t),\ j_i^{g}(t),\ \left\{ c_{i,k}^{g}(t) \right\}_{k \in \{G,L,H\}},\ \tau_i^{g}(t) \right\}_{t=gT}^{(g+1)T-1},$$

   where $o_i^{g}(t) \in \{S, N\}$ and $j_i^{g}(t) \in \{G, L, H\}$, subject to the per-period budget and time constraints.

2. Given prices, wages, and technologies, the representative producer in each sector $j \in \{G, L, H\}$ maximizes profits in every period $t \in \mathbb{Z}_{\geq 0}$ by choosing its demands for skilled and unskilled labor efficiency units, $S_j(t)$ and $N_j(t)$, respectively.

[16] The proof for (11) is located in Appendix B.3.

[17] Note that if $\lambda_j = 1$, (11) collapses to a homothetic demand function.

3. All sectoral output markets and skilled and unskilled labor markets clear in every period $t \in \mathbb{Z}_{\geq 0}$.

3.3 Market Clearing

Structural Change. The Roy-type selection problem together with the assumption of free labor mobility across sectors for skilled and unskilled workers implies the continued relocation across sectors until sectoral wages equalize. Accordingly, for a given occupation

$$w_{j,o}(t) = w_o(t), \qquad \forall j \in \{G, L, H\}, \quad o \in \{S, N\}. \tag{12}$$

Each sector therefore faces a common wage rate for skilled and unskilled efficiency units of labor. Since each sector operates in a perfectly competitive environment, the unit cost of output is the price of one unit of output $p_j(t)$ in the given sector. After replacing the conditional demands for skilled and unskilled labor into the CES aggregator (8)[18], I obtain an expression for the price of one unit of output in a given sector j

$$p_j(t) = \left(\phi_j^{\epsilon} w_N(t)^{1-\epsilon} + \left(1 - \phi_j\right)^{\epsilon} \left(\frac{w_S(t)}{B(t)} \right)^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}} \frac{1}{A_j(t)}. \tag{13}$$

Equation (13) writes the price of one unit of output in sector j as the inverse of TFP multiplied by the unit cost of composite labor input which is a function of wages, skill intensity and skill-biased technical change. That is, a higher skill intensity $1 - \phi_j$ implies greater sensitivity to the wage rate of skilled labor. Conversely, faster skill-biased technical change makes skilled labor units more productive, effectively reducing the unit cost of skilled labor. Finally, a higher level of TFP in sector j implies greater efficiency in producing output in that sector, naturally reducing both the unit cost and the price of output in that sector. [19]

Human Capital Allocation. Likewise, the Roy-type selection problem together with the assumption of free labor mobility across occupations implies that a parent with a human capital level h_i^g will pursue skilled work if $h_i^g \geq \bar{h}(t)$ where $\bar{h}(t)$ is constructed using the indifference condition

$$\bar{h}(t) = \frac{w_N(t)}{w_S(t)}. \tag{14}$$

[18] See the rest of Appendix B.1 for the formal proof.

[19] An implication of having $p_H(t)$ as the numéraire is that conditional on knowing the wage rate for one unit of unskilled labor, $w_N(t)$, one can pin down the wage rate for one unit of skilled labor, $w_S(t)$, using (13) and then pin down both $p_L(t)$ and $p_G(t)$ using the same equation.

Note that the indifference condition does not depend on time allocation thanks to Proposition 1. The threshold (14) indicates that in addition to the the raw human capital level of the parent, the supply of skills in the economy is also pinned down by general equilibrium effects.

Market Clearing. Recall that $\mu(h, h'; t)$ denotes the joint probability density function at time t of a generic parent-child human capital pair $h_i^g = h$ and $h_i^{g+1}(t) = h'$ and that $\mathcal{L}$ represents the (fixed) mass of dynasties in this economy. The sectoral goods market clearing conditions can be written as

$$A_j(t)Z_j(t) = \mathcal{L} \int_h \int_{h'} c_j(h, h'; t)\mu(h, h'; t)dh'dh \quad \forall j \in \{G, L, H\}, \tag{15}$$

where the LHS is the supply and the RHS is the demand for a given sector's output. Meanwhile, the market clearing conditions for skilled and unskilled labor can be written as

$$\sum_{j\in\{G,L,H\}} S_j^d(t) = \mathcal{L} \int_{\bar{h}(t)}^{\infty} \int_{h'} h\,(1 - \tau(h, h'; t))\,\mu(h, h'; t)dh'dh \quad \text{and} \tag{16}$$

$$\sum_{j\in\{G,L,H\}} N_j^d(t) = \mathcal{L} \int_0^{\bar{h}(t)} \int_{h'} (1 - \tau(h, h'; t))\,\mu(h, h'; t)dh'dh, \tag{17}$$

where the LHS is the sum of skilled and unskilled efficiency units of labor demanded by the three firms while the RHS is the time-adjusted supply of skilled and unskilled efficiency units of labor, respectively.

3.4 Steady State

Sections 3.2 and 3.3 characterize the equilibrium conditions that must hold in the transitional dynamics of this economy. Motivated by the growing importance of High-Skill Services with development, the remainder of this section studies the properties of an economy where all activity is occupied by the H-sector.

Proposition 2 (Asymptotic Dominance of the H-sector)**.** *Suppose that* $\lambda_H > \lambda_L, \lambda_G$, $g_L, g_G > g_H > 0$, $0 < \sigma < 1$ *and* $0 < \phi_j < 1$. *The parameter restrictions are sufficient for asymptotic dominance of the H-sector expenditure share.*

Proof. See Appendix B.4 for the formal proof. □

Proposition 2 establishes the conditions under which as $t \to \infty$, $e_{i,H}^g(t) \to 1$ and $e_{i,L}^g(t), e_{i,G}^g(t) \to 0$. This proposition is used to close the model; the implied limiting case of the proposition is that the H-sector is the only active sector in the economy. Accordingly,

this economy asymptotically hires only skilled workers and economy-wide TFP now grows at the constant growth rate g_H. Let $\tilde{x}(t)$ denote variable x in a steady state period $t \geq \mathcal{T}$. The representative firm's profit-maximizing equilibrium condition is

$$\tilde{w}_H(t) = \tilde{p}_H(t)\tilde{A}_H(t). \tag{18}$$

With $\tilde{p}_H(t)$ as the numéraire, this pins down the equilibrium wage for skilled workers

$$\tilde{w}_H(t) = \tilde{A}_H(t). \tag{19}$$

To make the model stationary in steady state, I proceed in two steps. First, I change the timing index t to s to denote the stage of childhood development. This step acknowledges that while the generation problem is stationary, the behavior of parents — and therefore aggregate and equilibrium objects — may vary over the stages of childhood development. Second, I normalize steady state consumption and output by $\tilde{A}_H(s)$ to obtain $\tilde{c}^{g*}_{i,H}(s)$ and $\tilde{Y}^{*}_{H}(s)$, respectively. The stationarized version of the labor market clearing condition can be written as

$$\tilde{S}^{d}_{H}(s) = \mathcal{L} \int_h \int_{h'} h \left(1 - \tilde{\tau}(h, h'; s)\right) \tilde{\mu}(h, h'; s) dh' dh, \tag{20}$$

where the LHS is the demand for skilled labor and the RHS is the time-adjusted aggregate supply of skills in the economy in steady state stage s of childhood development. The corresponding sectoral good market clearing condition can be written as

$$\tilde{S}^{s}_{H}(s) = \mathcal{L} \int_h \int_{h'} \tilde{c}^{*}_{H}(h, h'; s) \tilde{\mu}(h, h'; s) dh' dh, \tag{21}$$

where the LHS is the normalized supply of H-sector output (i.e. $\frac{\tilde{Y}_H(s)}{\tilde{A}_H(s)}$) and the RHS is the normalized aggregate demand for H-sector output.

4. Mapping the Model to Data

This section discusses how the model is taken to the data. It starts by briefly discussing some assumptions underlying the numerical solution. The remainder of this section lays out the calibration strategy, model fit and validation. The list of parameters to be calibrated is $\{\beta, \beta_g, \sigma, \chi, \epsilon, A_H(0), A_L(0), A_G(0), B(0), g_H, g_L, g_G, g_B, \phi_H, \phi_L, \phi_G, \theta_H, \theta_L, \theta_G, \lambda_H, \lambda_L, \lambda_G, \omega, \eta_c, \kappa_c\}$ together with the stage-specific human capital technology parameters $\{\zeta_s\}_{s=1}^{S}$. The parameters are disciplined using a combination of external calibration, direct measurement and internal calibration.

4.1 Numerical Solution

A detailed explanation of the solution algorithm is relegated to Appendix C.1. Instead, I elaborate here on two key assumptions that are required to proceed with the numerical solution. The first concerns the initial parent human capital distribution. This characterizes the model's initial conditions and is required to shoot forwards for the joint distribution. I assume that, like the initial child human capital distribution, the initial parent human capital distribution is governed by a lognormal distribution $G(\eta, \kappa)$ where η and κ are the location and shape parameters, respectively.

The second key assumption is around timing. I assume that there are three stages of childhood development (i.e. $S = 3$) and that each stage of childhood development lasts for a single time period in the model such that ($S = T = 3$). As per the model's demographic structure, this means that each individual from a given generation lives $2T = 6$ periods in total, 3 as a child and 3 as a parent. This assumption is invoked without loss of generality and purely for computational tractability. Empirically, I handle this by assuming that a single time period is equivalent to 5 years in the data. Furthermore, I follow Del Boca, Flinn and Wiswall (2014) in assuming that the functional form for the stage-specific parameters that shape the human capital technology is

$$\zeta_s = \exp\left(\xi_0 + \xi_1 \left(s - 1\right)\right), \qquad s = \{1, 2, 3\}. \tag{22}$$

This means that the human capital technology parameters are effectively disciplined by the pair $\{\xi_0, \xi_1\}$.

4.2 External Calibration

Of the 31 parameters to be calibrated, I externally calibrate the following 13 parameters $\{\beta, \beta_g, \sigma, \omega, \chi, \epsilon, A_H(0), A_L(0), A_G(0), B(0), g_H, g_L, g_H\}$. Before proceeding, it is necessary to comment on the interaction between TFP and skill-biased technical change. US sectoral growth rates in technical change can be computed from US KLEMS (Jorgenson, Ho and Samuels, 2017) and its extension (Elbadri and Vizcaino, 2025). Assuming that the true production function is homogeneous of degree 1 as implied by CES aggregator (7), then the change in technical change that is measured from the data is

$$\begin{aligned} \Delta \ln \hat{A}_j(t) &= \Delta \ln Y_j(t) - \Omega_{j,N}(t) \Delta \ln N_j(t) - \Omega_{j,S}(t) \Delta \ln S_j(t) \\ &= \Delta \ln A_j(t) + \Omega_{j,S}(t) \Delta \ln B(t) \quad \text{where} \quad \Omega_{j,N}(t) + \Omega_{j,S}(t) = 1. \end{aligned} \tag{23}$$

That is, observed growth in measured TFP $\Delta \ln \hat{A}_j(t)$ contains both skill-neutral TFP growth $\Delta \ln A_j(t)$ (which maps directly to the model's $A_j(t)$) and a (sector-neutral) skill-biased technical change growth component $\Omega_{j,S}(t)\Delta \ln B(t)$ where $\Omega_{j,S}(t)$ and $\Omega_{j,N}(t)$ are the cost shares of skilled and unskilled labor in sector j, respectively. To recognize this, I introduce new notation. I replace the calibrated parameters $A_j(0)$ and g_j that represent the sector-specific, skill-neutral TFP initial level and growth rate with $\hat{A}_j(0)$ and $\hat{g}_j$ which denote the initial TFP level and growth rate that is measured from the data and therefore contaminated with skill-biased technical change. I then net out skill-biased technical change from the measured TFP levels using a model-based decomposition that simulates the cost shares. Incorporating this change means that the vector of parameters that I calibrate either through external calibration or direct measurement is now $\{\beta, \beta_g, \sigma, \omega, \chi, \epsilon, \hat{A}_H(0), \hat{A}_L(0), \hat{A}_G(0), B(0), \hat{g}_H, \hat{g}_L, \hat{g}_G\}$.

Table 1 presents the values and sources of the externally calibrated parameters. The period discount factor β is assigned a value of 0.904. As in Barro and Sala-i-Martin (2004), this assumes an annual discount factor of 0.98. Since a period in the model is 5 years, the discount factor is retrieved by 0.98^5. I use an inter-generational discount factor β_g of 0.5 which is assumed by Boar and Lashkari (2021). Together, these values suggest that parents hardly attach value to the continuation value of their grandchildren. The elasticity of substitution of 0.1 suggests strong complementarities in consumption between the different sectoral goods which is a common finding in the structural change literature (Herrendorf, Rogerson and Valentinyi, 2013; Buera and Kaboski, 2009). I use the Frisch elasticity value of 0.5 from Boppart, Krusell and Olsson (2024) to inform the value of χ and normalize the disutility constant ω to unity.

The elasticity of substitution between skilled and unskilled workers in the CES aggregator (8) is assigned a value of 2.9 implying gross substitutability between both types of labor. This value is obtained by Acemoglu and Autor (2011) who fit the method by Katz and Murphy (1992) to longer US time series data. Sectoral TFP in the model grows exogenously according to an exponential growth function. All sectoral (measured) TFP and skill-biased technical change initial levels are normalized to unity in the initial period. The growth rates of sectoral TFP observed in the data (that are contaminated with skill-biased technical change) are obtained from the 44-industry version of the Elbadri and Vizcaino (2025) dataset which runs from 1947 to 2023. These are calculated from the industry-level growth rates and aggregated using Törnqvist weights. The industry classification into broad sectors is consistent with Appendix A.7. The growth rates in Table 1 are per period (i.e. per 5 calendar years) and, unsurprisingly, show that the High-Skill Services sector TFP grows at the slowest rate, followed by the Low-Skill Services sector sector.

Table 1. Externally Calibrated Parameters

Parameter	Description	Value	Source
Panel A. Preferences			
β	Period discount factor	0.904	Barro and Sala-i-Martin (2004)
β_g	Intergenerational discount factor	0.500	Boar and Lashkari (2021)
σ	Elasticity of substitution across sectoral goods	0.100	Buera and Kaboski (2009)
χ	Frisch elasticity	0.500	Boppart, Krusell and Olsson (2024)
ω	Work disutilty constant	1.000	Normalization
Panel B. Technology			
ϵ	Elasticity of substitution between skilled and unskilled labor, L-sector	2.900	Acemoglu and Autor (2011)
$\hat{A}_H(0)$	Initial measured TFP, H-sector	1.000	Normalization
$\hat{A}_L(0)$	Initial measured TFP, L-sector	1.000	Normalization
$\hat{A}_G(0)$	Initial measured TFP, G-sector	1.000	Normalization
$B(0)$	Initial skill-biased technical change level	1.000	Normalization
$\hat{g}_H$	Average measured TFP growth per period, H-sector	0.053	Elbadri and Vizcaino (2025)
$\hat{g}_L$	Average measured TFP growth per period, L-sector	0.005	Elbadri and Vizcaino (2025)
$\hat{g}_G$	Average measured TFP growth per period, G-sector	0.037	Elbadri and Vizcaino (2025)

Notes: This table reports parameters calibrated outside the model using external evidence. One period corresponds to 5 years. The H-, L- and G-sectors denote the High-Skill Services, Low-Skill Services and Goods sectors, respectively.

4.3 Internal Calibration

The remaining 17 parameters are calibrated internally. After normalizing λ_L to 1 and making use of the property that expenditure shares sum to 1 (which means that $\theta_L = 1 - \theta_G - \theta_H$), the remaining 15 parameters make up the vector of free parameters left to calibrate

$$\Theta \equiv \{\phi_H, \phi_L, \phi_G, g_B, \theta_G, \theta_H, \lambda_G, \lambda_H, \xi_0, \xi_1, \zeta_3, \nu, \eta, \kappa, \eta_c, \kappa_c\}.$$

The vector Θ is jointly identified to match patterns of structural change, time use and the skill distribution in postwar United States. The candidate vector that is chosen is the one that yields the minimum discrepancy between model-simulated and empirical data moments. This is accomplished via simulated method of moments (SMM).

At any given time, the model has just one cohort of parents and one cohort of children. Reconciling cross-sectional data with model therefore requires building and tracking synthetic cohorts over time. I operationalize this by partitioning the sample period (1942-

present) into six synthetic cohorts. This means that each calendar year is mapped to a unique combination of parent generation and childhood stage of development, as per the timing assumptions of the model. Each parent generation-child stage period lasts for a five-year window. More details on the timing convention can be found in Appendix C.2 and the full list of moments that I explicitly target together with details on data sources and construction are listed in Appendix C.3.

The choice of data moments is made such that the model reproduces patterns of structural change, time allocation and the skill distribution of the postwar US economy while simultaneously ensuring that the Jacobian matrix is of full rank. Heuristically, one can see this by splitting the parameters and moments into three blocks. The first block is concerned with matching the evolution of structural change in the US. The parameters that belong in this block are $\{\theta_H, \theta_G, \lambda_H, \lambda_G, \phi_H, \phi_L, \phi_G, g_B\}$. The consumption technology weights $\{\theta_G, \theta_H\}$ can be mapped to the initial aggregate expenditure shares[20][21] while the parameters $\{\lambda_G, \lambda_H\}$ are particularly helpful in shaping current expenditure shares given initial shares by virtue of the fact that these two parameters control the income elasticity of demand. The sector-specific skill intensities $\{\phi_H, \phi_L, \phi_G\}$ are used to match the initial cost share of skilled workers (relative to the skilled and unskilled wage bill) in each sector. Finally, the growth rate in skilled-biased technical change g_B is informative for the direction and magnitude of the price of Goods relative to High-Skill Services given that sectors vary in skill intensity.

The second block is concerned with matching time use patterns. The parameters that belong in this block are $\{\xi_0, \xi_1, \nu\}$. The parameters $\{\xi_0, \xi_1\}$ control the importance of parental childcare time allocation in the human capital law of motion in each stage of childhood development as per (22), and therefore shape the continuation value that parents face. Cross-stage variation in childcare time allocation obtained from stage-parental childcare time allocation profiles can pin down stage-specific elasticities. Simultaneously, ν shapes the weight and curvature of parental time allocation to childcare in flow utility, thereby determining whether, and the extent that, income effect dominates the substitution effect. This can be pinned down by the trend of parental time allocated to the childcare of young children over generations.

The final block targets the supply of skills in the US. The parameters $\{\eta, \kappa, \eta_c, \kappa_c\}$ are placed in this block. The initial parent human capital distribution follows a lognormal

[20]While employment shares could have substituted for expenditure shares as a measure of structural change, I choose the former because it maps directly to the household's demand system.

[21]Since the model defines output as value added, as intermediate inputs are not explicitly modeled, the consumption expenditure shares must be computed in value-added terms using the Total Requirements matrix from the Input–Output tables Herrendorf, Rogerson and Valentinyi (2013). Appendix section C.4 provides a brief overview of the steps implemented to do this at a sectoral level.

distribution $G(\eta, \kappa)$ where η and κ are the location and shape parameters, respectively. Conditional on the occupational choice threshold (which is endogenous), η and κ control the share of skilled workers and the variance of earnings conditional on being skilled (proxied by the log 90-10 wage gap conditional on being a skilled worker) in the first generation, respectively. Likewise, fixing the human capital production function parameters and the occupational choice threshold, the lognormal initial child human capital draw distribution $G(\eta_c, \kappa_c)$ can be disciplined by targeting the same moments of the sixth generation (current) US skill distribution. Appendix C.5 reports, for each parameter, the most responsive moment, computed from the Jacobian of the moments with respect to the parameters.

Table 2 presents the values of the internally calibrated parameters. The SMM produces a higher Goods sector consumption aggregator weight relative to the other two sectors in order to match the higher expenditure share in that sector in initial periods. With gross complementarities in sectoral preferences ($\sigma < 1$), the income elasticity parameters show that the High-Skill Services sector is the income-elastic sector (since $\lambda_H > 1$) while the Goods sector is the income-inelastic sector (since $\lambda_L < 1$) relative to Low-Skill Services. Furthermore, I find that the High-Skill Services sector is the most skill-intensive sector while the Goods sector is the least skill-intensive sector, as in Fang and Herrendorf (2021) who use the same sector split. Finally, $g_B > 0$ confirms the presence of skill-biased technical change, with skilled workers benefiting disproportionately more than unskilled workers. Combined, the SMM results are fully aligned with the parameter restrictions required for asymptotic convergence to the High-Skill Services sector (Proposition 2).

Transitioning to Panel B, the time use parameters imply that $\zeta_1 = 0.255$, $\zeta_2 = 0.500$ and $\zeta_3 = 0.983$. Given that the elasticity on parental time allocation is $1 - \zeta_s$, this implies that parental time investments are declining in importance over the stages of childhood development. That is, the marginal benefit of one additional unit of time allocated to childcare is worth less for developing the child's human capital as the child gets older. This finding is closely aligned with the literature who also find that parental time investments are key in early stages of childhood development but fall with the child's age (Del Boca, Flinn and Wiswall, 2014; Francesconi and Heckman, 2016; Yum, 2023; Del Bono et al., 2016). The CRRA parameter's value exceeds unity which means that matching the trend in childcare time allocation requires that the income effect dominate the substitution effect such that childcare time allocation is perceived as a luxury relative to work. Thus, as households get richer, they bias their time allocation away from work and towards childcare while simultaneously biasing their consumption towards High-Skill Services.

Panel C reveals the skill distribution parameters that are necessary to match the initial parent and child human capital lognormal distributions. The location parameter is higher

Table 2. Internally Calibrated Parameters

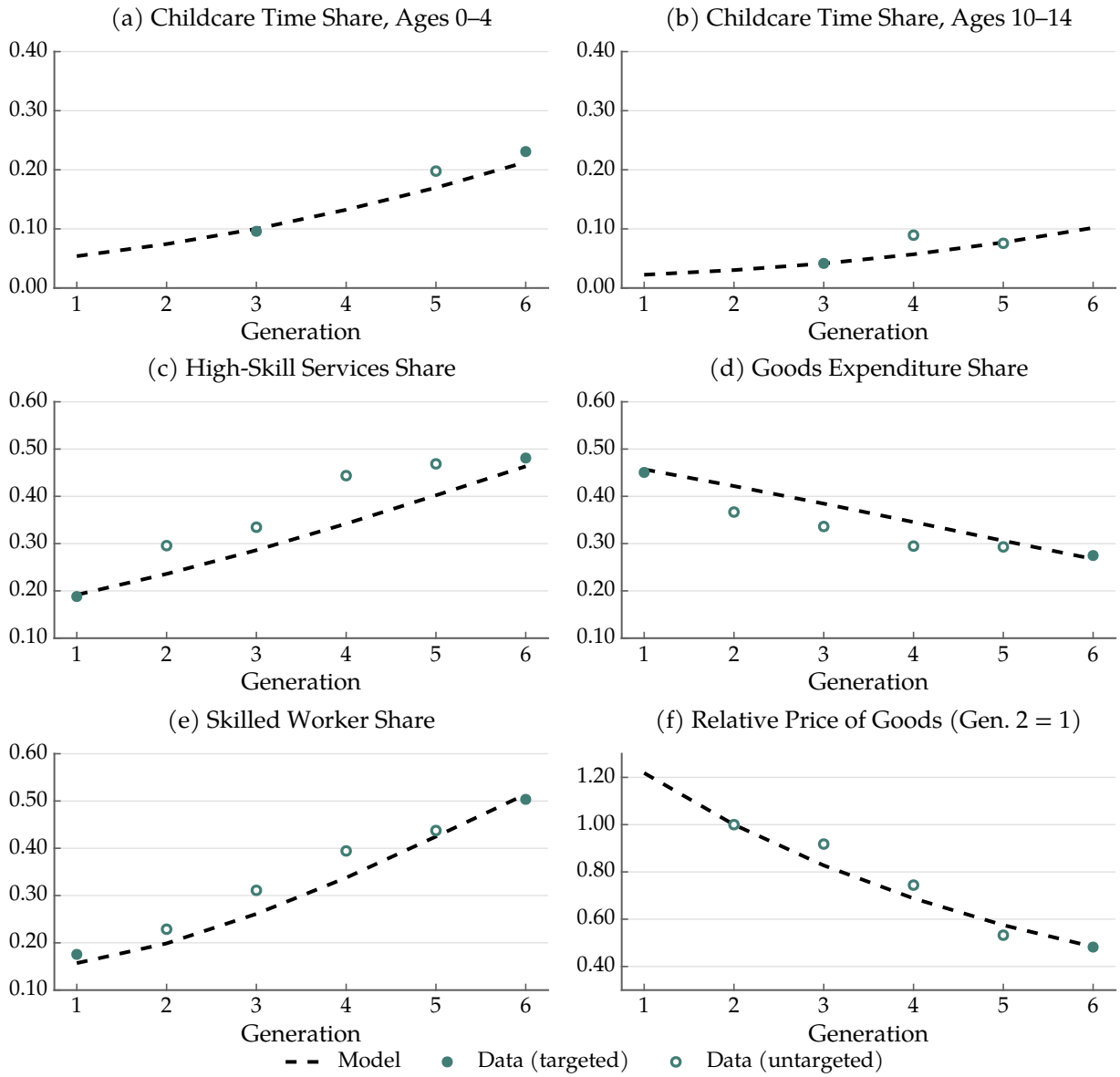
Parameter	Description	Value	Targeted moment
Panel A. Structural Change			
θ_G	Consumption weight, G-sector	0.433	G-sector expenditure share (Gen. 1)
θ_H	Consumption weight, H-sector	0.254	H-sector expenditure share (Gen. 1)
λ_G	Income elasticity parameter, G-sector	0.970	G-sector expenditure share (Gen. 6)
λ_H	Income elasticity parameter, H-sector	2.128	H-sector expenditure share (Gen. 6)
ϕ_H	Unskilled labor intensity, H-sector	0.358	Skilled share of H-sector wage bill (Gen. 1)
ϕ_L	Unskilled labor intensity, L-sector	0.472	Skilled share of L-sector wage bill (Gen. 1)
ϕ_G	Unskilled labor intensity, G-sector	0.508	Skilled share of G-sector wage bill (Gen. 1)
g_B	Growth rate of skill-biased technical change	0.103	Change in p_G/p_H (Gen. 2 to Gen. 6)
Panel B. Time Use			
ξ_0	Level, human capital technology	−1.367	Childcare time, stage 1 (Gen. 3)
ξ_1	Stage gradient, human capital technology	0.675	Childcare time, stage 3 (Gen. 3)
ν	CRRA parameter	1.943	Childcare time, stage 1 (Gen. 6)
Panel C. Skill Supply			
η	Location, initial parental distribution	−2.298	Skilled worker share (Gen. 1)
κ	Shape, initial parental distribution	1.414	90–10 log wage gap, skilled workers (Gen. 1)
η_c	Location, child endowment distribution	−3.445	Skilled worker share (Gen. 6)
κ_c	Shape, child endowment distribution	1.119	90–10 log wage gap, skilled workers (Gen. 6)
Panel D. Normalizations			
θ_L	Consumption weight, L-sector	0.313	Set to $1 - \theta_G - \theta_H$
λ_L	Income elasticity parameter, L-sector	1.000	Reference for λ_G, λ_H

Notes: The parameters in Panels A–C are jointly estimated by simulated method of moments (SMM) to match 15 data moments; the final column reports the moment each parameter is most closely associated with. Gen. g denotes generation g; skilled worker shares, expenditure shares, wage gaps and wage-bill shares are averages within each generation. The H-, L- and G-sectors denote the High-Skill Services, Low-Skill Services and Goods sectors, respectively.

for the initial parent human capital distribution compared to the child's, meaning that a non-negligible portion of parent human capital is accumulated during childhood rather than genetically. Panel D is implied by the calibrated parameters in Panel A.

Figure 6 shows the extent to which the model with the given parameter selection can replicate patterns of structural change, time allocation and the skill distribution in postwar United States. In all six panels, the x-axis is the generation. As per the timing convention in Appendix C.2, Generation 1 begins from 1942 and lasts until 1956 while Generation 6 lasts from 2017 to 2031. Solid circles are targeted data moments, hollow circles are untargeted data moments and the dashed lines are the model-simulated moments. Starting with parental time allocation, the model is able to replicate the cross-stage variation in both levels and changes in parental childcare time allocation (as a share of childcare and work time) over time as shown in Panels 6a and 6b. This is particularly helpful for disciplining the stage-specific parental time elasticities in the human capital production function. Looking at Panel 6c (6d) shows that the model can replicate the long-term rise (decline) in the High-Skill Services (Goods) sector expenditure share in the US. Panel 6e shows that the simulation closely tracks the observed rise in the share of skilled workers over time while Panel 6f shows that the model matches the decline in the relative price of Goods sector output relative to High-Skill Services. Overall, Figure 6 shows that the model can rationalize the rise in time allocated to childcare, High-Skill Services and the skilled worker share over time in postwar US.

Figure 6. Data vs. Model-Simulated Moments



Notes: Solid circles are data moments targeted in the SMM estimation; hollow circles are untargeted. The dashed line is the model simulation.

In addition to the variables in Figure 6, I also match the initial share of the sectoral wage bill allocated to skilled workers. These are used to infer the sectoral skill intensities. Table (3) reports the comparison between the data and model-simulated moments; the model is able to match the data within 5% for all three sectors.

Table 3. Skilled Share of the Sectoral Wage Bill: Model vs. Data

Sector	Data	Model	%Δ
High-Skill Services (H)	0.538	0.556	3.5
Low-Skill Services (L)	0.238	0.242	1.8
Goods (G)	0.170	0.174	2.4

Notes: The skilled share of the wage bill in sector j is $w_S S_j/(w_N N_j + w_S S_j)$, averaged over generation 1. %Δ is the model's deviation from the data, $100 \times$ (model − data)/data. All three moments are targeted in the SMM.

4.4 Model Validation

Thus far, I have shown that the model is capable of matching the patterns of structural change, time use and the skill distribution in postwar US. In this subsection, I validate the model by comparing how it performs against *untargeted* moments from US data. Table 4 shows how the model performs against the data on moments of interest that are not explicitly targeted.

Table 4. Untargeted Moments: Model vs. Data

Moment	Data	Model	%Δ	Source
Change in skill premium (Gen. 3 to Gen. 6)	1.227	1.290	5.1	Ruggles et al. (2025)
H-sector employment share (Gen. 1)	0.130	0.160	23.4	Ruggles et al. (2025)
H-sector employment share (Gen. 6)	0.425	0.385	−9.4	Ruggles et al. (2025)
Change in p_L/p_G (Gen. 2 to Gen. 5)	1.359	1.183	−13.0	Elbadri and Vizcaino (2025)

Notes: These moments are not targeted in the SMM. The skill premium is the mean hourly wage of skilled relative to unskilled workers. In the model, it is $w_S(t)\,\mathbb{E}[h \mid \text{skilled}](t)/w_N(t)$, where $w_S(t)$ is the wage per efficiency unit of skilled labor, $w_N(t)$ is the unskilled wage, and $\mathbb{E}[h \mid \text{skilled}](t)$ is the average human capital of skilled workers; in the data, it is the ratio of mean hourly wages of skilled and unskilled workers. %Δ is $100 \times$ (model − data)/data. H-sector denotes the High-Skill Services sector.

The wage premium — the ratio of the hourly wage of a skilled and unskilled worker — is a moment of interest that has received attention (Katz and Murphy, 1992; Buera et al., 2022). The data shows that the wage premium has risen by an average of 22.7% between generation 3 to 6.[22] The model is able to closely track this, only slightly over-predicting the wage premium over the same time period by 6.3 percentage points. Similarly, the model matches the trend in the High-Skill Services employment share, over-predicting the initial share and under-predicting the current share by 3 and 4 percentage points, respectively. This means that in the model, sectoral expenditure and employment shares are closely

[22]This corresponds to the change in the wage premium between today and the 1972-1986 average.

correlated. Finally, while the model generates a more muted rise in the price of output in Low-Skill Services relative to Goods, the model reproduces the decline in the Goods and Low-Skill Services price (in the correct order) relative to High-Skill Services.

5. Quantitative Exercises

This section performs three counterfactual exercises using the quantitative version of the model. The first exercise quantifies the role of initial conditions — TFP in particular — in shaping structural change. The second and third exercises focus on quantifying the effect of early childhood development on structural change. Specifically, the second exercise assesses the role of the rise in parental time allocation observed in the US while the third exercise assesses the role of dynamic complementarity in shaping human capital outcomes and, by extension, structural change.

5.1 Exercise 1: Initial TFP and Structural Change

In the first counterfactual exercise, I ask to what extent differences in productivity levels can generate the cross-country differences in structural change, time use and early childhood human capital documented in Section 2. This implies changing the primary lever of economic development in the model — the initial sector-neutral TFP level. I experiment with both an increase and a decrease in the initial (generation 1) TFP level by 25% which is broadly equivalent to the US-Zimbabwe and US-South Africa TFP gap in the 1960s according to the Penn World Tables (Feenstra, Inklaar and Timmer, 2015).[23] All other parameters are fixed to their baseline calibration values.

The results of this counterfactual are reported in Table 5. The first column reports the outcome of interest. The remaining columns report the values of the corresponding outcome with an *initial* TFP level that is 25% below, equal to, and 25% above the baseline value, respectively, in generations 1 and 6. Panel A shows that, when comparing the same generation, the average time allocation of parents towards the childcare of young children (ages 0–4) rises with the TFP level. Panel B shows that average early childhood human capital also rises with the TFP level in the same generation, as does the share of skilled workers in the economy. Panel C shows that the High-Skill Services expenditure and employment shares are all also rising with TFP when comparing the same generation. The immediate observation from Table 5 is that the model can rationalize cross-country

[23] This 25 percentage point gap is in the observed (measured) TFP. This corresponds to $\ln \hat{A}_j(t)$ in equation (23) and is therefore contaminated by skill-biased technical change. In this counterfactual exercise, I fix the level and path of skill-biased technical change (initially normalized to unity) in both experiments. I alter the initial TFP level by 25% both ways, implying that both $A_j(0)$ and $\hat{A}_j(0)$ vary by 25% from baseline (that is, in each sector by the same magnitude). The growth rate of $A_j(t)$ in all sectors is the same as in baseline.

differences in time allocation, the supply of skills and, ultimately, the sectoral composition of aggregate output and employment documented in Section 2. [24]

Table 5. Initial TFP Levels and Cross-Country Patterns

	$\hat{A}_j(0) = 0.75$		Baseline		$\hat{A}_j(0) = 1.25$	
Outcome	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
Panel A: Time Use						
Childcare time share, ages 0–4	0.049	0.188	0.054	0.213	0.060	0.238
Childcare time share, ages 10–14	0.018	0.074	0.022	0.102	0.027	0.128
Panel B: Skill Formation						
Human capital of 4-year-olds (baseline = 1)	0.980	0.906	1.000	1.000	1.024	1.091
Share of skilled workers	0.153	0.484	0.157	0.514	0.161	0.544
Panel C: Structural Change						
H-sector expenditure share	0.158	0.410	0.192	0.463	0.221	0.504
H-sector employment share	0.130	0.334	0.160	0.385	0.186	0.426

Notes: The table reports the results of the first counterfactual exercise which alters the initial measured TFP in all three sectors, $\hat{A}_j(0)$, by both 25% either way. The H-sector refers to the High-Skill Services sector.

A closer look at Table 5 reveals that differences in initial productivity do not just lead to differences in the supply of skills and structural change, but also that these differences compound over time. This is particularly clear from panels B and C. Starting with panel B, differences in the average human capital of 4-year-old children are within 3% either way when comparing the higher and lower initial TFP experiments with the baseline in generation 1. In generation 6, however, these differences are noticeably magnified to fall within a 10% range either way. Similarly, the share of skilled workers is within half a percentage point either way in generation 1, rising to 3 percentage points either way in generation 6.

Several forces drive this amplification result. First, the initial parent and child human capital distributions are the same across all experiments, meaning that the generation 1 differences in the share of skilled workers are driven purely by general equilibrium effects that change the occupational choice threshold. However, the strength of income effects that affect the time allocation problem that parents face now varies with the initial TFP level. The same parent in the low-TFP experiment allocates less time to childcare than in

[24]This reinforces the finding from Restuccia and Vandenbroucke (2014) that productivity can explain cross-country differences in human capital.

the high-TFP experiment because their income is now less. This explains why the human capital of 4-year-old children rises with initial TFP.

By Generation 6, this force is amplified for several reasons. First, the supply of skills in generation 2 is now higher in the high-TFP case relative to the low-TFP case because of greater time investments by generation 1 parents. This compounds, eventually magnifying differences in the supply of skills by generation 6. Generation 6 parents now not only benefit from higher income because of benefiting from higher human capital *and* TFP levels, but they also become more productive at childcare which enters the child's human capital production function (6). This increases the marginal benefit of parental time investments which in turn is another source of motivation towards greater childcare allocated to young children. With higher early childhood human capital in the high-TFP case relative to the low-TFP case in generation 6, the level effect of higher early childhood human capital is compounded by dynamic complementarity which induces further investments in children's human capital as they get older. Indeed, this is underscored by the proportionally larger differences in the childcare time share towards *older* children in generation 6 relative to generation 1 across the different experiments. Overall, this amplifies differences in the share of skilled workers by Generation 6 across the different experiments.

Both supply- and demand-side dynamics operate in tandem to generate large cross-country disparities in the sectoral composition of output. Panel C shows that differences in the High-Skill Services expenditure and employment shares across the different initial TFP levels rise from about 3 percentage points either way in generation 1 to about 4–5 percentage points either way in generation 6. Aside from the human capital dynamics, sectoral demand is non-homothetic meaning that as incomes rise, households disproportionately increase demand for High-Skill Services. As such, supply- and demand-side dynamics are not just jointly determined, but they also reinforce each other, amplifying cross-country differences in human capital and structural change with initial productivity.

5.2 Exercise 2: Parental Time Investments and Structural Change

The second and third counterfactual exercises aim to quantify the contribution of early childhood development on structural change. In the second exercise, I do this by calculating the effect of the rise in US parental time allocated to young children on structural change. This is motivated by the steep rise in the time allocated to the childcare of young children by adults over time across countries and over time in the US as well as the comparatively larger elasticity of parental time elasticity in the multi-stage human capital accumulation technology. This is done by freezing the time allocated to the childcare of young children to the model-simulated generation 1 average of 5.4%. Parents now only optimize their time allocation when children exceed 4 years of age.

Table 6 reports the results of this counterfactual exercise. The first column reports the outcome of interest. The subsequent two columns report the baseline values for generations 1 and 6. In this counterfactual exercise, I distinguish between partial and general equilibrium, where the former fixes relative prices to their baseline paths while the latter allows these to clear markets. The subsequent two columns report the results under partial equilibrium while the final two columns report the results under general equilibrium. The headline result is reported in Panel D which shows that the High-Skill Services expenditure (employment) share in the general equilibrium counterfactual achieves just 19.0 (17.1) of the 27.2 (22.5) percentage point increase in the baseline between generation 1 and 6, implying that the rise in parental childcare time allocation towards young children in the US accounts for roughly a quarter to a third of structural change to High-Skill Services.

Table 6. Parental Time Investments in Childcare and Structural Change

	Baseline		PE		GE	
Outcome	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
Panel A: Time Use						
Childcare time share, ages 0–4	0.054	0.213	0.054	0.054	0.054	0.054
Childcare time share, ages 10–14	0.022	0.102	0.021	0.036	0.021	0.047
Panel B: Skill Formation						
Share of skilled workers	0.157	0.514	0.157	0.270	0.151	0.430
Skilled work threshold (baseline Gen. 1 = 1)	1.000	0.596	1.000	0.596	1.041	0.436
Mean human capital | skilled (baseline Gen. 1 = 1)	1.000	0.871	1.000	0.445	1.028	0.367
Panel C: Structural Change						
H-sector expenditure share	0.192	0.463	0.202	0.367	0.198	0.388
H-sector employment share	0.160	0.385	0.168	0.231	0.160	0.331
H-sector output (baseline Gen. 1 = 1)	1.000	2.843	1.082	1.454	1.068	1.382
p_G/p_H (baseline Gen. 1 = 1)	1.000	0.396	1.000	0.396	1.016	0.354
p_L/p_H (baseline Gen. 1 = 1)	1.000	0.525	1.000	0.525	1.013	0.484
Panel D: Change from Gen. 1 to Gen. 6						
Δ H-sector expenditure share	0.272		0.165		0.190	
Δ H-sector employment share	0.225		0.063		0.171	

Notes: The table reports the results of the second counterfactual exercise, which quantifies the effect of the rise in parental time allocated to the childcare of young children (ages 0–4) on structural change. The H-, L- and G-sectors denote the High-Skill Services, Low-Skill Services and Goods sectors, respectively. p_H, p_L and p_G denote the price of one unit of High-Skill Services, Low-Skill Services and Goods sector output, respectively. PE and GE denote partial and general equilibrium, respectively.

Starting with the partial equilibrium result, the childcare time share towards young children (ages 0–4) is frozen in all generations at the baseline generation 1 average. Fewer parental time investments when children are youngest (where the elasticity is highest) across generations mechanically reduces the human capital of parents in subsequent generations relative to the baseline. Dynamic complementarity and less parent human capital mute incentives to invest time in children's human capital in subsequent stages of childhood development relative to the baseline. These effects compound, explaining the sharp decline in the generation 6 childcare time share towards older children (ages 10–14) relative to the baseline. This in turn reduces the supply of skills *despite* the wage rate premium (and hence the skilled work threshold) being unchanged, as evidenced by both the decline in the share of skilled workers in the economy and the mean human capital level of skilled workers, which almost halves in generation 6 relative to the baseline. Put differently, the distribution of human capital has effectively shifted to the left (i.e. less human capital across the board), causing a sharp decline in the High-Skill Services employment share in generation 6 relative to the baseline since fewer workers now cross the threshold for skilled work. Simultaneously, the demand for High-Skill Services declines because households are now poorer due to less human capital and, given non-homothetic preferences, spend a smaller share of their income on High-Skill Services.

General equilibrium effects dampen the partial equilibrium effect of parental time investments in the childcare of young children on structural change. In general equilibrium, relative prices and wages clear markets. Accordingly, agents respond to the lower supply of human capital resulting from fewer time investments by parents over time. The wage rate premium rises in response to the decline in the supply of skills in the economy, reducing the threshold to accessing skilled work from 0.596 in the generation 6 baseline to 0.436 in the general equilibrium counterfactual and offsetting some of the losses in time investments in partial equilibrium as children age. This allows less skilled parents to access skilled work, allowing the share of skilled workers to recover from 27.0% in the generation 6 partial equilibrium to 43.0% in general equilibrium. Equivalently, the relative price of High-Skill Services rises in response to the decline in the supply of skills in the economy, effectively offsetting the sharp decline in the quantity of High-Skill Services demanded and produced in the economy. In general equilibrium, both the High-Skill Services expenditure and employment shares therefore recover part of the losses observed in the partial equilibrium result. Indeed, the quantity of High-Skill Services produced in generation 6 declines most in general equilibrium relative to partial equilibrium and baseline, indicating that the slight recovery in the High-Skill Services expenditure share masks a sharper drop in output more than offset by a rising relative price of the sector. Overall, these results indicate that the

rise in the childcare time allocated by parents towards young children accounts for up to a third of the rise of the High-Skill Services share.

5.3 Exercise 3: Dynamic Complementarity and Structural Change

Incorporating the multi-stage human capital production function (6) into a benchmark structural change model is one of the innovations of this paper. A central feature of this technology is that it embeds the notion of dynamic complementarity: a higher level of early childhood human capital raises the productivity of investments in subsequent stages of childhood development. The cumulative nature of the human capital accumulation process makes a strong case for the importance of early childhood development in explaining the supply of skills in the economy and, ultimately, the sectoral composition of aggregate output. The third counterfactual exercise quantifies how important dynamic complementarity is in shaping human capital investment decisions and, by extension, structural change in the US. This exercise is implemented by modifying the human capital production function (6). The human capital production will only feature the human capital draw of a child at birth (i.e. the genetic component of human capital, modeled as $h_i^{g+1}(gT)$ in all stages of childhood development rather than the current human capital of the child, implying the following setup

$$h_i^{g+1}(t+1) - h_i^{g+1}(t) = h_i^{g+1}(gT)^{\zeta_s+q}\left(h_i^g \tau_i^g(t)\right)^{1-\zeta_s}, \tag{24}$$

such that the cross-partial derivative $\frac{\partial^2 h_i^{g+1}(t+1)}{\partial h_i^{g+1}(t)\partial \tau_i^g(t)} = 0 \forall s = 1, 2$. Intuitively, (24) suggests that the human capital that a child has acquired in their early childhood no longer shapes parental time investment decisions in subsequent stages of childhood development. The former freezes relative prices at baseline values while the latter allows relative prices and wages to clear markets.

The results of the third exercise are presented in Table 7. The dynamics closely follow those in the second counterfactual exercise. The first column reports the outcome of interest. The next two columns report the baseline values for generations 1 and 6. The following two columns report the results under partial equilibrium. The final two columns report the results under general equilibrium. The headline result is reported in Panel D which shows that the High-Skill Services expenditure (employment) share in the general equilibrium counterfactual achieves just 17.8 (15.8) of the 27.2 (22.5) percentage point increase in the baseline between generation 1 and 6. Put differently, dynamic complementarity from past investments in human capital explain 34.6% and 29.8% of structural change in expenditure and employment to the High-Skill Services sector.

Table 7. Dynamic Complementarity and Structural Change

	Baseline		PE		GE	
Outcome	Gen. 1	Gen. 6	Gen. 1	Gen. 6	Gen. 1	Gen. 6
Panel A: Time Use						
Childcare time share, ages 0–4	0.054	0.213	0.029	0.033	0.033	0.073
Childcare time share, ages 10–14	0.022	0.102	0.019	0.022	0.019	0.030
Panel B: Skill Formation						
Share of skilled workers	0.157	0.514	0.157	0.156	0.153	0.341
Skilled work threshold (baseline Gen. 1 = 1)	1.000	0.596	1.000	0.596	1.022	0.346
Mean human capital | skilled (baseline Gen. 1 = 1)	1.000	0.871	1.000	0.427	1.015	0.300
Panel C: Structural Change						
H-sector expenditure share	0.192	0.463	0.197	0.349	0.194	0.372
H-sector employment share	0.160	0.385	0.164	0.184	0.159	0.317
H-sector output (baseline Gen. 1 = 1)	1.000	2.843	1.057	1.286	1.044	1.049
p_G/p_H (baseline Gen. 1 = 1)	1.000	0.396	1.000	0.396	1.008	0.322
p_L/p_H (baseline Gen. 1 = 1)	1.000	0.525	1.000	0.525	1.007	0.451
Panel D: Change from Gen. 1 to Gen. 6						
Δ H-sector expenditure share	0.272		0.152		0.178	
Δ H-sector employment share	0.225		0.020		0.158	

Notes: The table reports the results of the third counterfactual exercise, which quantifies the role of dynamic complementarity in the human capital production function for structural change. The H-, L- and G-sectors denote the High-Skill Services, Low-Skill Services and Goods sectors, respectively. p_H, p_L and p_G denote the price of one unit of High-Skill Services, Low-Skill Services and Goods sector output, respectively. PE and GE denote partial and general equilibrium, respectively.

Starting with the partial equilibrium result, parental time allocated to childcare when children are youngest falls *endogenously* relative to the baseline because while these investments mechanically raise the human capital of the child, parents internalize that this will no longer raise the marginal benefit from parental time investments in subsequent stages of childhood development. Consequently, parental time investments in the third stage of childhood development (i.e. ages 10–14) also substantially decline in both level and trend. While the skill price per efficiency unit (and, by extension, the skilled work threshold) is fixed to baseline levels, fewer parental time investments reduce the supply of skills relative to the baseline, evident from the decline in the share of skilled workers from 51.4% in generation 6 in the baseline to 15.6% in the partial equilibrium counterfactual. Furthermore, many of the skilled workers now suffer from lower human capital due to the

decline in parental time investments, effectively halving the mean human capital of skilled workers. Without any price adjustments, structural change measured by the change in the High-Skill Services expenditure and employment shares declines by 44.1% and 91.1% in partial equilibrium, respectively.

In general equilibrium, the decline in the supply of skills raises the relative price of High-Skill Services. The rise in the skill price incentivizes a slight recovery in human capital and, together with the declining occupational choice threshold for skilled work, the share of skilled workers recovers to 34.1% in generation 6. Overall, however, this reduces the quality of skilled workers even more than in partial equilibrium, with the average human capital of a skilled worker in generation 6 declining by 70.0% relative to generation 1 of the baseline in the general equilibrium counterfactual, compared to 57.3% in partial equilibrium and just 12.9% in the baseline. This is because skilled work is now accessible to less capable parents. This suggests that, in general equilibrium, dynamic complementarity affects the human capital quality margin even more than the quantity margin. Both the expenditure and employment shares of High-Skill Services rise due to these general equilibrium effects, which mitigate the sharp decline in both observed in the partial equilibrium counterfactual. As in the second counterfactual exercise, the 2.5 percentage point larger rise in the High-Skill Services expenditure share between generations 1 and 6 in general equilibrium relative to partial equilibrium masks a sharper decline in the physical output of High-Skill Services, offset by the rise in the sector's relative price. Nevertheless, the results suggest that dynamic complementarity explains approximately a third of structural change (measured in both expenditure and employment) to High-Skill Services.

6. Paid Parental Leave as an Industrial Policy

The quantitative exercises confirm that rising early childhood human capital has had a non-negligible contribution to the sectoral reallocation of economic activity in favor of High-Skill Services in postwar US and shed light on the mechanisms that drive this. A natural extension is therefore to consider what role, if any, policy can play in shaping structural change. That is, can childcare policies serve as a form of *industrial policy* that can influence the long-term composition of aggregate output?

6.1 Design and Implementation

I experiment with paid parental leave policies which involve covering a fixed share of foregone wages to parents at a given stage of childhood development using a tax implemented to all parents such that the budget balances in each period. Paid parental leave is a natural policy experiment for two reasons. First, parental time investments are

financed by the household through foregone earnings, meaning that the policy can alter the time allocation tradeoffs that parents face when children are youngest. Second, because this is a common policy in advanced economies. For example, Germany's *Elterngeld* program is a state-funded parental allowance that replaces two-thirds of the earnings parents forego to care for a newborn, for up to twelve months and subject to a cap. Sweden and Norway operate similar schemes, which are financed through social insurance contributions on earnings.

The benchmark paid parental leave policy design that I consider is the following. A government can cover a fixed portion of gross income (i.e. before adjusting for time allocated to childcare) that the parent loses from their time allocated to the childcare of their young child (i.e. in the first stage of childhood development, ages 0–4) for up to a cap on time allocated to childcare. That is, the parent will not accrue any extra paid leave from allocating more time than the cap to childcare. This policy applies only to parents with a young child (ages 0–4) and is removed when children exceed this age range.

To implement this numerically, two changes to the benchmark model in Section 3 are required. The first is to the household budget constraint (5) which now becomes

$$\left(1 - T^P(t)\right)\left(1 - \tau_i^g(t) + b(t)\min\{\tau_i^g(t), L\}\right) y_i^g(t) \geq p_i^g(t) c_i^g(t). \tag{25}$$

$T^P(t)$ is the flat tax rate that the government charges at time t on the gross income (earnings plus benefits) of parents with a stage-1 child, so that $\left(1 - T^P(t)\right) y_i^g(t)$ is the post-tax income the household would earn before making its time allocation choice.[25] $b(t)$ is the share of forgone gross earnings, and hence also of forgone after-tax earnings, that the household recovers for time spent with its child. The term $\min\{\tau_i^g(t), L\}$ caps the benefit at L, the maximum share of the stage's time for which a parent receives it: if $\tau_i^g(t) > L$, the parent receives a benefit of $b(t)L\, y_i^g(t)$.

The second change introduces a government that implements this policy. The government sets the tax rate $T^P(t)$, collects the taxes and uses them to finance the benefit paid to households. With h denoting parent human capital, h' child human capital and $\mu(h, h'; t)$ their joint distribution among parents with a stage-1 child, the government budget constraint is

[25] Proposition 1, which ensures that the parent finds it optimal to maximize pre-tax, pre-time-adjusted income, continues to hold because the tax rate is flat, independent of human capital, and therefore scales income proportionally with $y_i^g(t)$.

$$\underbrace{\mathcal{L}\int_{h}\int_{h'} T^{P}(t)\left(1-\tau(h,h';t)+b(t)\min\{\tau(h,h';t),L\}\right)y(h;t)\,\mu(h,h';t)\,dh\,dh'}_{\text{government revenue}} =$$

$$\underbrace{\mathcal{L}\int_{h}\int_{h'} b(t)\min\{\tau(h,h';t),L\}\,y(h;t)\,\mu(h,h';t)\,dh\,dh'}_{\text{government expenditure}}. \quad (26)$$

The LHS aggregates taxes collected from parents which form government revenue while the RHS aggregates the paid benefit paid out to parents which form government expenditure. The government budget constraint (26) imposes an additional market clearing condition; the tax rate $T^{P}(t)$ is chosen by the government to clear its budget constraint conditional on the policy parameters $\{b(t), L\}$ which are set exogenously. This policy experiment is active only when children are in the first stage of childhood development and is otherwise inactive.

6.2 Results

The policy experiment sets $b = 2/3$ and $L = 0.2$. This means that the government transfers each parent 2/3 of their forgone after-tax earnings from childcare with a cap of 20% of their time. These parameters are chosen to match Germany's *Elterngeld* scheme which pays 65-67% of forgone earnings for one year (which matches 20% of the model's 5-year periods). The tax rate clears the government's budget constraint, and this policy only applies when parents have a child in the first stage of childhood development (i.e. ages 0–4).

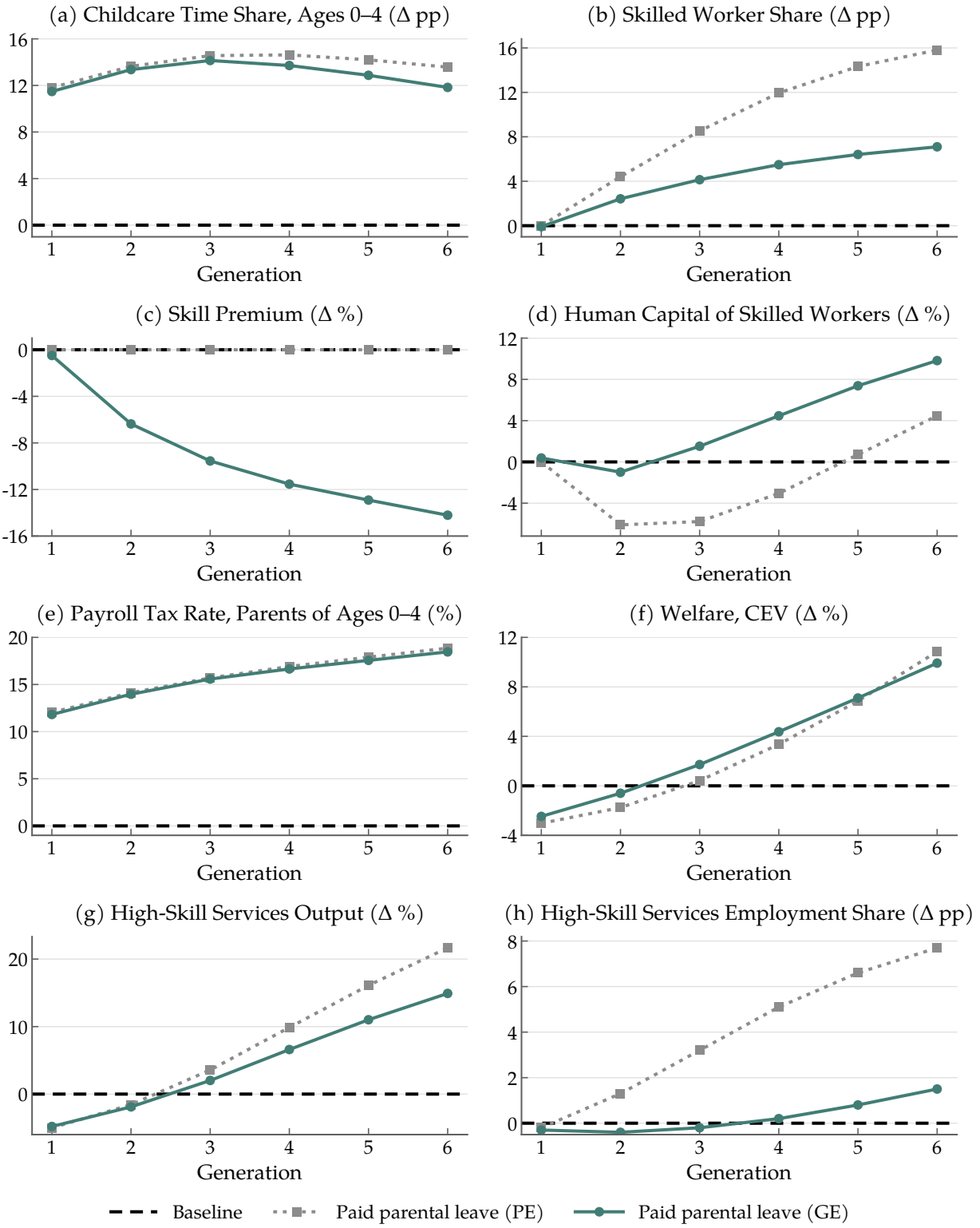
The results of this policy experiment are presented in Figure 7. As with the counterfactual exercises, I distinguish between partial and general equilibrium, where the former freezes wages and prices at the no-policy baseline values while the latter allows them to clear markets. In all Figure 7 panels, the variable of interest is the change relative to the no-policy baseline in the corresponding generation either in percentage points (for share variables) or percent (for level variables). Accelerations are computed as the change between generations 1 and 6 under the policy relative to the same change in the baseline. I begin with partial equilibrium. The time share allocated to the childcare of young children is higher than in the no-policy counterfactual, suggesting that the policy does indeed alter parents' time allocation trade-off in favor of childcare. Through dynamic complementarity, this leads to a sustained divergence in the share of skilled workers. In the initial generation, the distribution of human capital is unchanged, but the policy leads to greater human capital investments by parents over time which drives the sustained amplification of gains in

the skilled workers share. The wage premium is unchanged relative to the baseline by construction and, as such, these gains represent a shift in the skill distribution rather than the occupational choice threshold. The policy has a u-shaped effect on the average human capital of skilled workers relative to the no-policy baseline. This is because the policy has made skilled work more accessible to future generations that would not have otherwise found it optimal to access skilled work in the no-policy baseline. These compositional effects fade by generation 5, and the average human capital of skilled workers is higher thereafter compared to the no-policy baseline. The payroll tax rate that clears the government's budget constraint rises from 12.0% in generation 1 to 18.9% in generation 6. This is because over time, the policy has raised the supply of skills and by extension household incomes, leading to a greater desire to allocate time to childcare and therefore take-up of the policy.

The welfare effects are captured by the ex-ante consumption-equivalent variation (CEV): the percent rise in consumption that is required from the corresponding point in time forever, for the parent and all descendants, that would make a parent, before knowing their own and their child's human capital, indifferent between the baseline and the policy. In partial equilibrium, the CEV in generation 1 from the introduction of the tax rate without any change in the generation's human capital distribution is -3.0%, meaning that welfare relative to the no-policy baseline is initially reduced. Over time, this reverses: by generation 3, the welfare effects of the policy are positive, and by generation 6 they are 10.9% implying that the policy is equivalent to offering parents in generation 6 and all their descendants 10.9% of extra consumption. In partial equilibrium, the physical quantity of High-Skill Services falls in generation 1 relative to the no-policy counterfactual but, like CEV, rises over time. The policy initially reduces labor supply in generation 1 as parents spend more time with their children, but by generation 3, the greater supply of skills reverses the effect of the policy on skilled labor supply. Indeed, the greater supply of skills in the economy accelerates the rise in the High-Skill Services employment share over time in partial equilibrium, with the policy causing an additional 7.9 percentage point increase in the sector's employment share between generations 1 and 6 relative to the baseline, which amounts to a 35.1% faster rise.

In general equilibrium, relative prices and wages are allowed to clear markets, and this dampens the effect of the policy. The key difference with the partial equilibrium exercise is that the wage rate premium now declines as the economy responds to a greater supply of skills. This in turn increases the occupational choice threshold, so that the rise in the skilled worker share is 20.1% faster than in the baseline, compared with 44.3% faster in partial equilibrium. The share of skilled workers and the average human capital of skilled workers nevertheless still see noticeable gains, and the payroll tax rate that clears the government budget is largely unchanged in partial and general equilibrium.

Figure 7. Paid Parental Leave as a Policy Experiment



Notes: Each panel plots the paid parental leave economy minus the baseline, by generation, in percentage points (pp) or percent (%). The benchmark policy is $b = 2/3$ and a cap $L = 0.2$. It is financed by a flat payroll tax on parents of children aged 0–4. The solid line is the general equilibrium (GE); the dotted line is partial equilibrium (PE) where prices and wages are held at their baseline paths. The dashed line marks the baseline.

Welfare effects are remarkably similar in both the partial and general equilibrium policy experiments. Indeed, the physical quantity of High-Skill Services output still benefits in the long-run, rising 25.6% faster than in the no-policy baseline between generations 1 and 6 (36.1% in partial equilibrium), so that by generation 6 it exceeds the baseline by 14.9% even when wages and prices dampen the effects of the policy. The High-Skill Services employment share rises only 8.0% faster than in the baseline, compared with 35.1% faster in partial equilibrium, even though the supply of skills has risen dramatically. The disconnect in the magnitude of the rise in High-Skill Services output and employment share in general equilibrium is due to the fact that the former mainly captures a rising skill intensity of each parent working in the High-Skill Services sector, while the latter captures the physical number of parents in the High-Skill Services sector, which rises by comparatively less because the sectors are strong complements.

Overall, the key takeaway from this policy experiment is that paid parental leave can lead to a sharp increase in childcare time allocation and, consequently, the supply of skills in the economy in the long run. Paid parental leave accelerates the rise in the supply of skills by a fifth and in High-Skill Services output by a quarter, but the decline in the relative price of High-Skill Services limits the acceleration of its employment share to 8.0%. Despite this, the welfare effects from paid parental leave are substantial.

6.3 Robustness

Next, I present robustness checks for the paid parental leave policy. This involves experimenting with different values of either b or L at any given point. I experiment with the values of 1 and 1/3 for b and 0.1 and 0.3 for L. Under each counterfactual, the government implements the flat payroll tax rate that balances the government budget. Table 8 reports the results of the robustness checks. The key outcomes of interest are the CEV, the quantity of High-Skill Services output, the High-Skill Services employment share and the payroll tax rate in generation 6. All figures are reported as the difference from their generation 6 counterparts in the no-policy baseline.

Table 8 presents the results of the robustness checks. Starting with panel A, I find that rises in the replacement rate raise welfare and the pace of structural change, but do so at a decreasing rate. Doubling b from 1/3 to 2/3 raises the CEV by 5.4 percentage points and the rise in both High-Skill Services output and employment share by 6.5 and 0.6 percentage points, respectively. Increasing b from 2/3 to 1, on the other hand, generates an additional 2.1 percentage point rise in the CEV and adds 4.7 and 0.4 percentage points to the increase in High-Skill Services output and employment share. This is because the leave cap binds for nearly all parents of young children by generation 6, so a higher replacement rate mainly raises the transfer on time parents would spend with their children anyway, rather than

Table 8. Paid Parental Leave – Robustness to the Replacement Rate and the Leave Cap

	CEV (%)	H-sector output (Δ %)	H-sector employment share (Δ pp)	Payroll tax rate (%)
Panel A: Replacement rate b (cap $L = 0.2$)				
$b = 1/3$	4.57	8.4	0.9	8.89
$b = 2/3$ (benchmark)	9.92	14.9	1.5	18.46
$b = 1$	12.02	19.6	1.9	25.80
Panel B: Leave cap L (replacement rate $b = 2/3$)				
$L = 0.1$	4.55	7.1	0.7	9.70
$L = 0.2$ (benchmark)	9.92	14.9	1.5	18.46
$L = 0.3$	15.24	23.6	2.3	26.13

Notes: The table reports general equilibrium outcomes in generation 6 under alternative paid parental leave designs relative to the no-policy baseline. H-sector denotes the High-Skill Services sector.

the return to additional childcare time at the margin. When comparing the leave cap, on the other hand, the rise in welfare and the acceleration of structural change coming from a rising L is roughly linear.

An important observation from Table 8 is that both combinations $\{b = 1, L = 0.2\}$ and $\{b = 2/3, L = 0.3\}$ require roughly the same payroll tax rate in generation 6 and are therefore about equally expensive. Despite this, the latter strictly dominates the former, with the rise in welfare being larger in magnitude and the pace of structural change accelerating. The key takeaway from this is that a policy that is less generous but lasts longer is more beneficial as a form of industrial policy and, ultimately, welfare.

7. Conclusion

This paper proposes a theory whereby early childhood development shapes structural change. I motivate my theory with empirical evidence in Section 2 that the High-Skill Services share, parental time allocated to younger children and early childhood human capital outcomes all rise with development. In Section 3, I build a multi-sector heterogeneous agents model in general equilibrium to with the aim of quantifying early childhood development's contribution to structural change and study policy experiments. In the model, parents are altruistic and enjoy spending time with their children. As sectoral TFP rises, non-homothetic sectoral preferences generate an increased demand for the income-elastic sector (High-Skill Services). Simultaneously, income effects and the rising wage premium induce parents to invest more time in their children in early stages of their childhood, where returns to human capital investments are highest. Through dynamic complementarity, this increases the supply of skilled workers in the economy which benefits the High-Skill Services sector disproportionately given that this sector employs skills more

intensively. Endogenizing supply- and demand-side forces means that the model can account for the feedback effects between both forces.

After calibrating the model to match patterns of structural change, time use and the skill distribution over time in the US in Section 4, I use the quantitative version of the model to perform three counterfactual exercises in Section 5. The first asks whether cross-country TFP disparities can rationalize the cross-country differences in time allocation, the supply of skills and structural change presented in Section 2. I find that indeed, differences in initial TFP of a magnitude resembling the US-Zimbabwe and US-South Africa TFP gaps in the 1960s generates differences in parental time allocation, early childhood human capital and structural change that are amplified over time. The second exercise asks how much the rise in parental time allocated to the childcare of young children has contributed to structural change in the US. I find that without the rise in parental time allocated to the childcare of young children observed in the data, the High-Skill Services employment and expenditure shares would have risen only by 17.1 and 19.0 percentage points compared to the 22.5 and 27.2 percentage points observed in the data, respectively. The third exercise quantifies the role of dynamic complementarity in shaping structural change to High-Skill Services in the US. The headline result is that without dynamic complementarity from past investments embedded in the multi-stage human capital technology, the High-Skill Services employment and expenditure shares would have risen by 15.8 and 17.8 percentage points relative to the observed 22.5 and 27.2 percentage points. This suggests that dynamic complementarity explains a third of US structural change to the High-Skill Services sector.

With early childhood development shaping the sectoral composition of economic activity, I proceed in Section 6 by asking whether paid parental leave can serve as a form of industrial policy. Specifically, I study a policy whereby the government levies a flat payroll tax that finances paid leave benefits. These benefits allow parents to recover a portion of the earnings forgone by spending time with their newborns. Numerically, the policy is parameterized to match the key features of Germany's state-financed paid leave program, *Elterngeld*. I find that the policy accelerates the rise in the High-Skill Services employment share by 8.0% over seven decades. This is smaller than the acceleration in the rise of the physical quantity of High-Skill Services by 25.6% over the same time horizon. This mismatch results from general equilibrium effects which, due to the sharper increase in the supply of skills, reduces the wage premium and the relative price of the High-Skill Services sector. Furthermore, I find that for a similar payroll tax rate, a longer leave cap generates a larger increase in welfare than a higher replacement rate.

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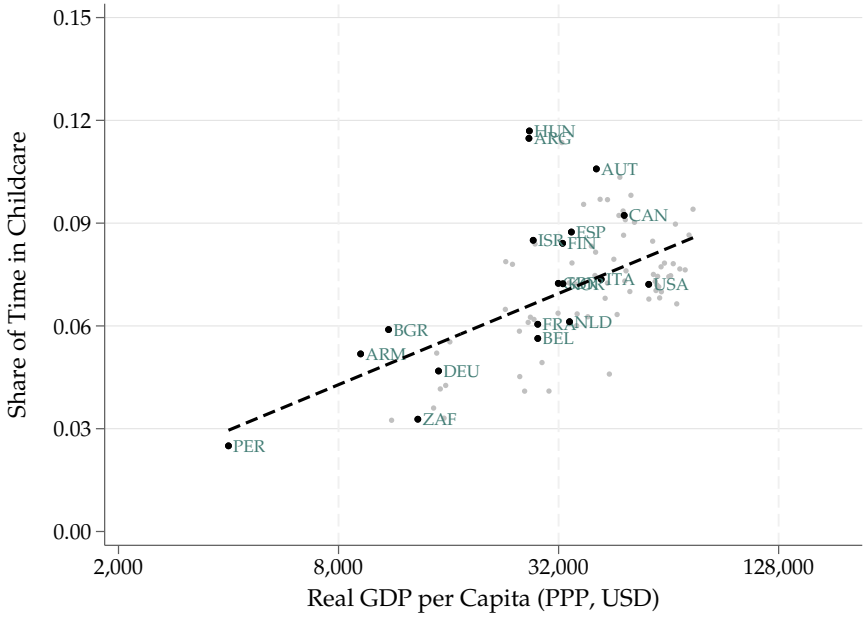
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A. Additional Empirics

A.1 Parental Time Investments in Childcare in Only-Child Households

Figure A.1 restricts attention to diarists who live in a household with a child below the age of 5 and that this is the only child in the household against PPP-adjusted real GDP per capita. The plot reveals that the trend in Figure 1 is robust in one-child households, reflecting a per-child trend.

Figure A.1. Share of Time Spent on Childcare (Only-Child Households)



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

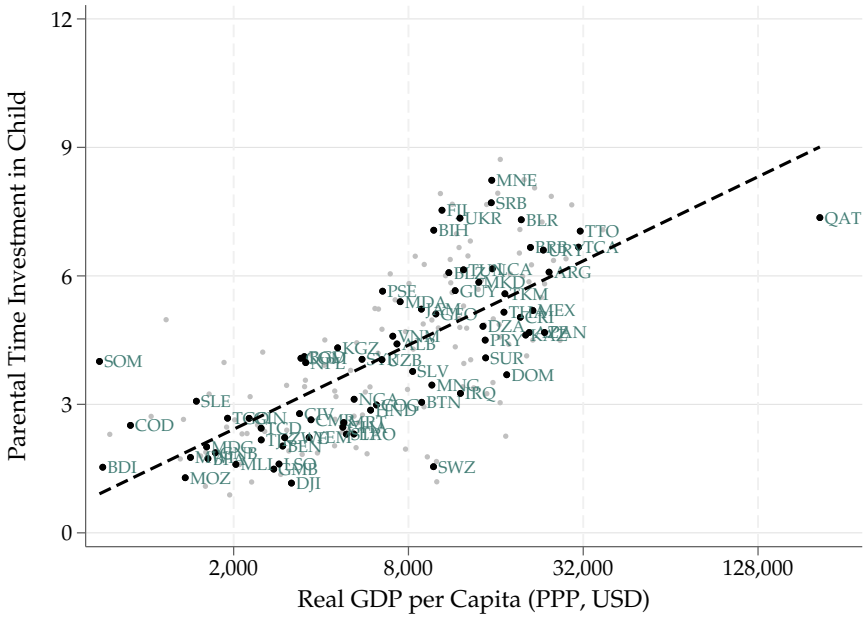
A.2 Parental Time Investments in Childcare Using IPUMS-MICS

The IPUMS-MICS dataset surveys the following six binary questions for mothers and fathers (separately) of children aged between 0 and 4 years old:

- Read books to the child in the last 3 days
- Told stories to child in the last 3 days
- Sang songs to the child in the last 3 days
- Took child outside in the last 3 days
- Played with the child in the last 3 days
- Spent time with child naming, counting, drawing things in the last 3 days

Aggregating the binary answers (1 for yes, 0 for no) for each household and then aggregating at the country-year level using the proposed survey weights yields the parental time investment in child index which is plotted against PPP-adjusted real GDP per capita in Figure A.2. It shows that the cross-country fact that parental time devoted to childcare rises with development even in the lower end of the development spectrum.

Figure A.2. Parental Time Investment in Childcare Index (Using IPUMS-MICS)

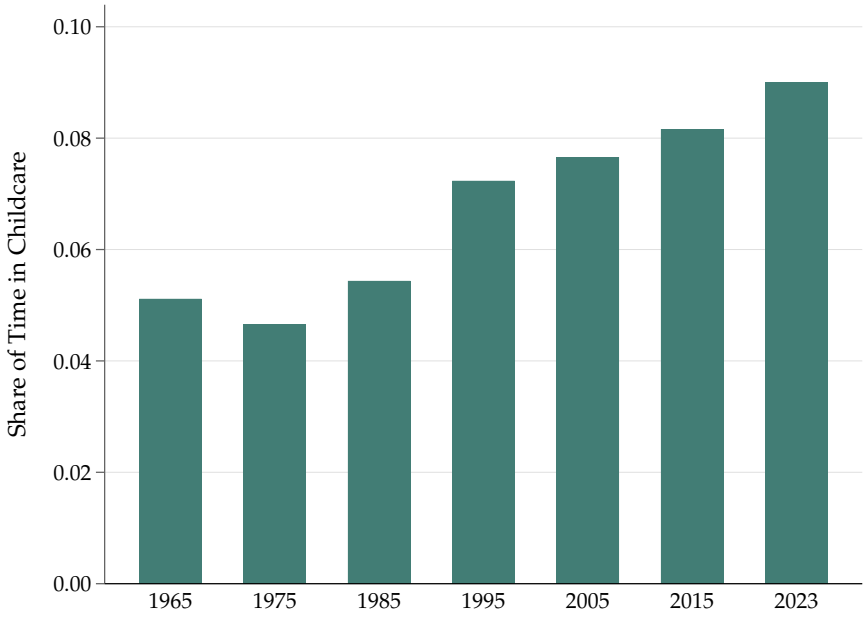


Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.3 Parental Time Investments in Childcare of Younger Children in US

Figure A.3 isolates US time use data from 1965 to 2023 shows that, with the exception of 1975, there is a consistent increase in time spent on childcare over time.

Figure A.3. Share of Time Spent on Childcare Over Time in the US

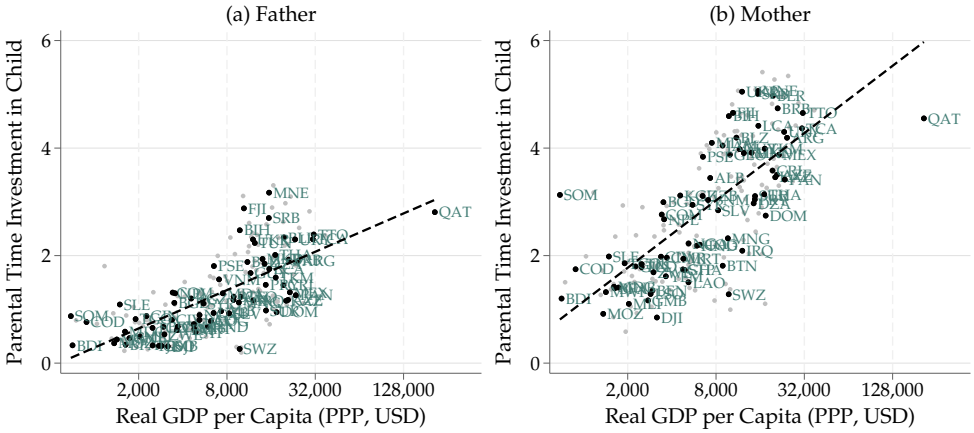


Notes: This chart restricts attention to US samples.

A.4 Parental Time Investments in Childcare of Younger Children By Gender Using IPUMS-MICS

Using IPUMS-MICS data and following the same strategy in Appendix A.2, I plot the parental time investment index by the gender of the parent in Figure A.4. The link shows a very similar trend to Figure 2; both parents spend more time with their children as countries get richer.

Figure A.4. Parental Time Investment in Childcare Index by Gender (Using IPUMS-MICS)

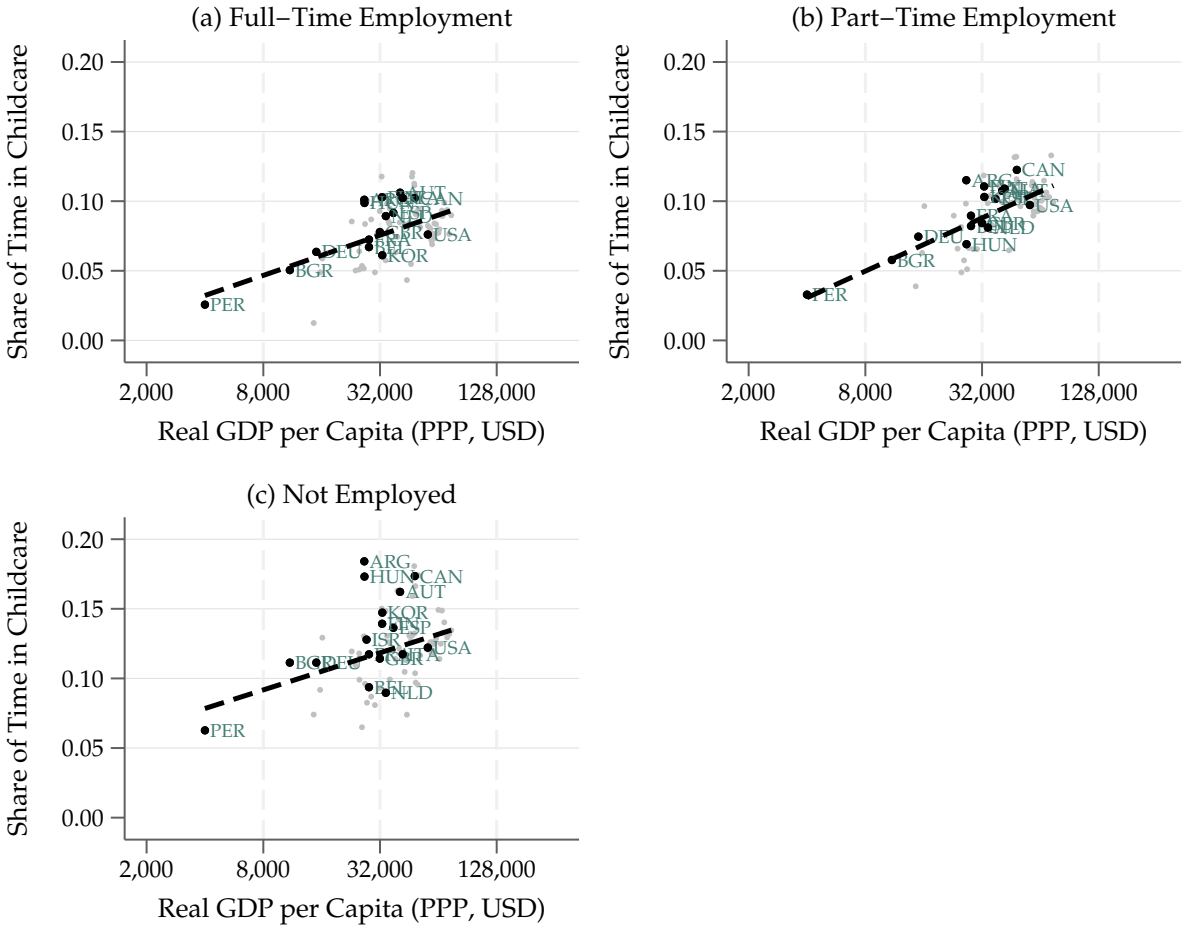


Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.5 Female Time Investments in Childcare of Younger Children By Employment Status

Using time use data, I plot the time allocated to the childcare of young children by females distinguished by employment status against PPP-adjusted real GDP per capita. Figure A.5 shows that the time allocation trend in Figure 2 holds regardless of employment status.

Figure A.5. Female Time Investment in Childcare by Employment Status

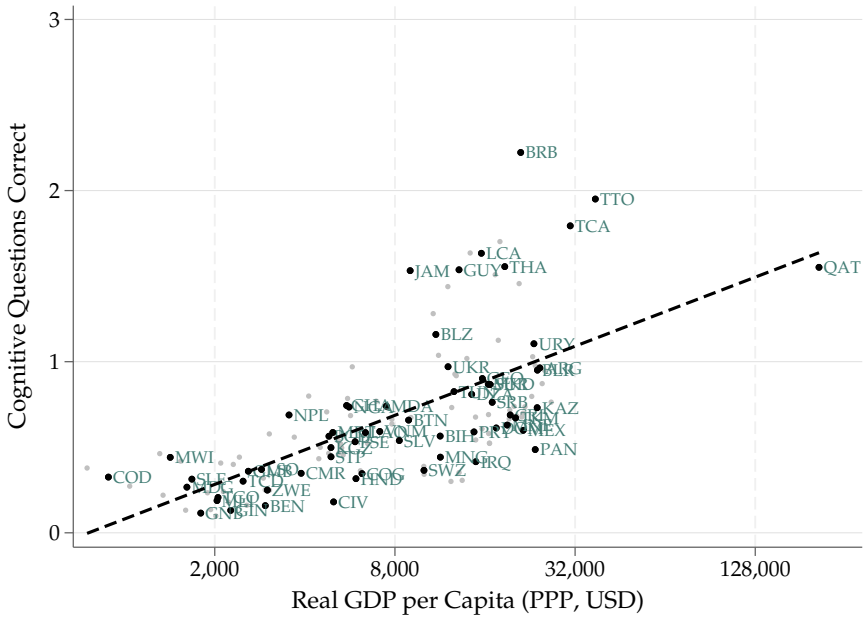


Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.6 Cognitive Skills of 3-Year-Old Children

Figure A.6 plots the average number of cognitive questions that 3-year-old children correctly answer. Similar to Figure 3 that plots the same variable for 4-year-old children, it shows that 3 year-old children in richer countries successfully answer more cognitive questions.

Figure A.6. Early Childhood Cognitive Development Demonstrated (3-Year-Olds)



Notes: Grey circles are country-year observations. Black circles are country averages. Country-year observations are aggregated from microdata using the proposed survey weights. The dashed line corresponds to the line of best fit for the grey circles.

A.7 Industry Classification

Table A.1 presents the industry classification for Services. The remaining industries make up the Goods sector. This includes Agriculture, Fishing and Forestry, Mining and Extraction, Manufacturing, Construction and Electricity, Gas, Water and Waste Management.

Table A.1. Skill Intensity of Service Industries, United States

Industry	1960	1970	1980	1990	2000	2005	2010	2015	2020	Average
Panel A. High-Skill Services										
Education	0.69	0.64	0.65	0.66	0.69	0.77	0.76	0.77	0.78	0.71
Health and social work	0.52	0.47	0.48	0.54	0.56	0.55	0.53	0.70	0.77	0.57
Financial services and insurance	0.40	0.40	0.45	0.55	0.59	0.69	0.72	0.53	0.59	0.55
Business services and real estate	0.46	0.44	0.49	0.50	0.52	0.54	0.53	0.63	0.68	0.53
Panel B. Low-Skill Services										
Other services	0.32	0.33	0.42	0.47	0.45	0.39	0.33	0.23	0.25	0.35
Public administration and defense	0.15	0.19	0.26	0.29	0.35	0.40	0.37	0.39	0.42	0.31
Wholesale and retail trade	0.20	0.17	0.22	0.26	0.27	0.29	0.29	0.13	0.15	0.22
Transport, storage and communications	0.10	0.10	0.14	0.17	0.18	0.14	0.29	0.28	0.28	0.19
Hotels and restaurants	0.17	0.14	0.15	0.17	0.14	0.20	0.19	0.13	0.14	0.16
Private household services	0.03	0.01	0.03	0.03	0.02	0.01	0.01	0.03	0.21	0.04

Notes: Each cell reports the person-weighted share of employment in an industry (IPUMS INDGEN) accounted for by skilled occupations. Skilled occupations are those classified under OCCISCO as legislators, senior officials and managers; professionals; or technicians and associate professionals. The sample includes employed individuals (EMPSTAT) aged 16–64. The final column is the average over 1960–2020.

B. Model Derivations

B.1 Firm problem

The j-sector producer selects the quantities of $S_j(t)$ and $N_j(t)$ to maximize profits given prices and wages

$$\max_{S_L(t),N_L(t)} \pi_L(t) = p_j(t)A_j(t)\left[\phi_j N_j(t)^{\frac{\epsilon-1}{\epsilon}} + \left(1-\phi_j\right)\left(B(t)S_j(t)\right)^{\frac{\epsilon-1}{\epsilon}}\right]^{\frac{\epsilon}{\epsilon-1}} - w_{L,S}(t)S_L(t) - w_{L,N}(t)N_L(t). \tag{B.1.1}$$

The first order conditions imply

$$w_{j,S}(t) = \left(1-\phi_j\right)p_j(t)A_j(t)\left(\frac{Z_j(t)}{S_j(t)}\right)^{\frac{1}{\epsilon}} B(t)^{\frac{\epsilon-1}{\epsilon}} \quad \text{and} \tag{B.1.2}$$

$$w_{j,N}(t) = \phi_j p_j(t)A_j(t)\left(\frac{Z_j(t)}{N_j(t)}\right)^{\frac{1}{\epsilon}}. \tag{B.1.3}$$

Taking the ratio of (B.1.2) and (B.1.3) and canceling out $Z_j(t)$ gives the relative quantities of skilled and unskilled labor in sector j as a function of wages, skill intensities and skill-biased productivity

$$\frac{S_j(t)}{N_j(t)} = \left(\frac{1-\phi_j}{\phi_j}\frac{w_{j,N}(t)}{w_{j,S}(t)}\right)^{\epsilon} B(t)^{\epsilon-1}. \tag{B.1.4}$$

A natural consequence of free labor mobility across sectors for both types of labor is that wages for each type of labor will equalize ($w_{j,o}(t) = w_o(t)$). To obtain price of output in sector j, I raise both (B.1.2) and (B.1.3) by $1-\epsilon$, multiply the former by ϕ_j^{ϵ} and the latter by $(1-\phi_j)^{\epsilon}$ to obtain

$$\phi_j^{\epsilon} w_{j,N}(t)^{1-\epsilon} = \phi_j\left(p_j(t)A_j(t)\right)^{1-\epsilon}\left(\frac{Z_j(t)}{N_j(t)}\right)^{\frac{1-\epsilon}{\epsilon}} \quad \text{and} \tag{B.1.5}$$

$$\left(1-\phi_j\right)^{\epsilon}\left(\frac{w_{j,S}(t)}{B(t)}\right)^{1-\epsilon} = \left(1-\phi_j\right)\left(p_j(t)A_j(t)\right)^{1-\epsilon}\left(\frac{Z_j(t)}{S_j(t)}\right)^{\frac{1-\epsilon}{\epsilon}} B(t)^{\frac{\epsilon-1}{\epsilon}}, \tag{B.1.6}$$

respectively. The next step is to sum up (B.1.5) and (B.1.6) where using the definition of $Z_j(t)$, I obtain

$$\phi_j^{\epsilon} w_{j,N}(t)^{1-\epsilon} + \left(1-\phi_j\right)^{\epsilon}\left(\frac{w_{j,S}(t)}{B(t)}\right)^{1-\epsilon} = \left(p_j(t)A_j(t)\right)^{1-\epsilon}. \tag{B.1.7}$$

Raising both sides by $\frac{1}{1-\epsilon}$, I obtain

$$p_j(t) = \left(\phi_j^{\epsilon} w_N(t)^{1-\epsilon} + \left(1 - \phi_j\right)^{\epsilon} \left(\frac{w_S(t)}{B(t)} \right)^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}} \frac{1}{A_j(t)}. \tag{B.1.8}$$

B.2 Proof for Proposition 1

I drop the generation superscript and dynasty subscript for expositional convenience. Assume a fixed parent human capital level, prices and wages. Let $c(X)$ denote real consumption when the parent spends X, defined by $X = p(c)\,c$; $c(X)$ is strictly increasing in X because expenditure is strictly increasing in real consumption. Let $V^*(y(t))$ denote the value of the parent's problem after maximizing utility conditional on gross income $y(t)$. Furthermore, let $\tau^*(t)$ denote the corresponding optimal time allocation to childcare with a gross income of $y(t)$. The parent's objective function is

$$V^*(y(t)) = \frac{c\left(y(t)(1-\tau^*(t))\right)^{1-\nu}}{1-\nu} - \omega\frac{(1-\tau^*(t))^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}} + \Phi(\tau^*(t)).$$

where $\Phi(\tau^*(t))$ captures the continuation value consisting of subsequent flow utilities and the present discounted value of future generations' utility under the optimal time allocation $\tau^*(t)$.

Consider now a higher gross income $y'(t) > y(t)$. The optimal time allocation $\tau^*(t)$ remains feasible and the continuation value $\Phi(\tau^*(t))$ is unchanged. Now, however, the parent can benefit from higher utility from consumption since

$$\frac{c\left(y'(t)(1-\tau^*(t))\right)^{1-\nu}}{1-\nu} - \omega\frac{(1-\tau^*(t))^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}} + \Phi(\tau^*(t)) > \frac{c\left(y(t)(1-\tau^*(t))\right)^{1-\nu}}{1-\nu} - \omega\frac{(1-\tau^*(t))^{1+\frac{1}{\chi}}}{1+\frac{1}{\chi}} + \Phi(\tau^*(t))$$

It follows that $V^*(y'(t)) > V^*(y(t))$. Accordingly, the parent's intertemporal utility is increasing in gross income.

B.3 Obtaining Marshallian Demands

Given optimal time allocation, each household chooses the consumption bundle that minimizes total expenditure on consumption given the non-homothetic CES aggregator $\forall t \in \left[gT, (g+1)T\right)$

$$\min_{\{c_{i,j}^g(t)\}_{t\in[gT,(g+1)T),\forall j\in\{G,L,H\}}} \sum_{j\in\{G,L,H\}} p_j(t)c_{i,j}^g(t) \quad \text{subject to} \quad \sum_{j\in\{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}}\right)^{\frac{\sigma-1}{\sigma}} = 1. \tag{B.3.1}$$

The corresponding Lagrangian is

$$\mathcal{L} = \sum_{j\in\{G,L,H\}} p_j(t)c_{i,j}^g(t) + \Lambda(t)\left[1 - \sum_{j\in\{G,L,H\}} \theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}}\right)^{\frac{\sigma-1}{\sigma}}\right], \tag{B.3.2}$$

where $\Lambda_j(t)$ is the Lagrange multiplier. The corresponding first order condition (FOC) is

$$p_j(t) = \Lambda(t)\frac{\sigma-1}{\sigma}\theta_G^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}}\right)^{\frac{-1}{\sigma}} \frac{1}{c_i^g(t)^{\lambda_j}}. \tag{B.3.3}$$

Multiplying both sides by the sectoral good $c_{i,j}^g$ implies

$$p_j(t)c_{i,j}^g(t) = \Lambda(t)\frac{\sigma-1}{\sigma}\theta_j^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}}\right)^{\frac{\sigma-1}{\sigma}}. \tag{B.3.4}$$

To get rid of the Lagrange multiplier, sum (B.3.4) over all j and plug in the non-homothetic CES aggregator

$$\sum_{j\in\{G,L,H\}} p_j(t)c_{i,j}^g(t) = \Lambda(t)\frac{\sigma-1}{\sigma}. \tag{B.3.5}$$

Defining $p_i^g(t)c_i^g(t) = \sum_{j\in\{G,L,H\}} p_j(t)c_{i,j}^g(t)$, (B.3.5) can be substituted into (B.3.3) to obtain

$$p_j(t) = p_i^g(t)c_i^g(t)\theta_G^{\frac{1}{\sigma}} \left(\frac{c_{i,j}^g(t)}{c_i^g(t)^{\lambda_j}}\right)^{\frac{-1}{\sigma}} \frac{1}{c_i^g(t)^{\lambda_j}}. \tag{B.3.6}$$

Rearranging (B.3.6) to solve for the sectoral good yields

$$c_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t) c_i^g(t)} \right)^{-\sigma} c_i^g(t)^{(1-\sigma)\lambda_j}. \tag{B.3.7}$$

While (B.3.7) is indeed the Marshallian demand as it expresses sectoral consumption in terms of total expenditure (which is income in equilibrium), prices and parameters, it can be re-expressed as a function of expenditure shares. To see this, multiply and divide both sides of (B.3.7) by the sectoral price $p_j(t)$ and by total expenditure $p_i^g(t)c_i^g(t)$ to obtain

$$\frac{p_j(t)c_{i,j}^g(t)}{p_i^g(t)c_i^g(t)} = \theta_j [p_i^g(t)c_i^g(t)]^{\sigma-1} p_j(t)^{1-\sigma} c_i^g(t)^{(1-\sigma)\lambda_j}. \tag{B.3.8}$$

Since $p_i^g(t)c_i^g(t) = \sum_{j\in\{G,L,H\}} p_j(t)c_{i,j}^g(t)$, it must be that $\sum_{j\in\{G,L,H\}} \frac{p_j(t)c_{i,j}^g(t)}{p_i^g(t)c_i^g(t)} = 1$ which allows (B.3.8) to take on the interpretation of a share. Accordingly,

$$c_{i,j}^g(t) = e_{i,j}^g(t) \frac{p_i^g(t)c_i^g(t)}{p_j(t)} \quad \text{where} \quad e_{i,j}^g(t) = \theta_j \left(\frac{p_j(t)}{p_i^g(t)} \right)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \tag{B.3.9}$$

To fully characterize (B.3.9), it is necessary to solve for the ideal price index $p_i^g(t)$. To pin down the price index, sum the expenditure share $e_{i,j}^g(t)$ over all j

$$1 = p_i^g(t)^{\sigma-1} \sum_{j\in\{G,L,H\}} \theta_j p_j(t)^{1-\sigma} c_i^g(t)^{(\sigma-1)(1-\lambda_j)}. \tag{B.3.10}$$

After re-arranging (B.3.10), the ideal price index $p_i^g(t)$ can be written as

$$p_i^g(t) = \left(\sum_{j\in\{G,L,H\}} \theta_j p_j(t)^{1-\sigma} c_i^g(t)^{(\lambda_j-1)(1-\sigma)} \right)^{\frac{1}{1-\sigma}}. \tag{B.3.11}$$

B.4 Proof of Proposition 2

Pre-requisites. To pin down the evolution of sectoral dynamics over time, two ingredients are necessary. The first is an expression for the evolution of relative sectoral expenditure shares of sector $k \in \{G, L\}$ in equilibrium. To obtain this, I start from the expenditure share equation (11),

$$e^g_{i,j}(t) = \theta_j \left(\frac{p_j(t)}{p^g_i(t)} \right)^{1-\sigma} c^g_i(t)^{(\sigma-1)\left(1-\lambda_j\right)}. \tag{B.4.1}$$

Taking the ratio of expenditure share of sector $k \in \{G, L\}$ over the expenditure share of sector H yields

$$R_k(t) = \frac{\theta_k p_k(t)^{1-\sigma} c^g_i(t)^{(\lambda_k-1)(1-\sigma)}}{\theta_H p_H(t)^{1-\sigma} c^g_i(t)^{(\lambda_H-1)(1-\sigma)}}. \tag{B.4.2}$$

Taking logs of (B.4.2), differentiating with respect to time and recognizing that $p_H(t)$ is the numeráire implies that

$$\frac{\dot{R_k}(t)}{R_k(t)} = (1-\sigma) \left(\frac{\dot{p_k}(t)}{p_k(t)} \right) + (\lambda_k - \lambda_H)(1-\sigma) \frac{\dot{c^g_i}(t)}{c^g_i(t)}. \tag{B.4.3}$$

Intuitively, (B.4.3) shows that the ratio of sectoral expenditure shares depends on how relative prices and income effects evolve. With gross complementarities in sectoral consumption (i.e. $\sigma < 1$), a decline in the relative price of sector k is one force in favor of the H-sector dominating as the k-sector would mechanically occupy a smaller share of expenditure. Another force is income effects: $\lambda_k < \lambda_H$ would imply that the H-sector is income elastic relative to the k-sector good, implying that the household biases consumption towards that sector provided that consumption growth is positive (i.e. $\frac{\dot{c^g_i}(t)}{c^g_i(t)} > 0$).

The second necessary ingredient is the evolution of the relative price of sector $k \in \{G, L\}$. To obtain this, I start from the expression for the price of output in a given sector (13),

$$p_j(t) = \left(\phi_j^{\epsilon} w_N(t)^{1-\epsilon} + \left(1-\phi_j\right)^{\epsilon} \left(\frac{w_S(t)}{B(t)} \right)^{1-\epsilon} \right)^{\frac{1}{1-\epsilon}} \frac{1}{A_j(t)}. \tag{B.4.4}$$

Taking the price of output in sector $k \in \{L, H\}$ relative to the numeráire $p_H(t)$ implies

$$p_k(t) = \frac{A_H(t)}{A_k(t)} \underbrace{\frac{\left(\phi_k^{\epsilon} + \left(1-\phi_k\right)^{\epsilon}\left(\frac{w_S(t)}{w_N(t)B(t)}\right)^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}}}{\left(\phi_H^{\epsilon} + \left(1-\phi_H\right)^{\epsilon}\left(\frac{w_S(t)}{w_N(t)B(t)}\right)^{1-\epsilon}\right)^{\frac{1}{1-\epsilon}}}}_{\Gamma_k(t)}. \tag{B.4.5}$$

Taking logs of (B.4.5) and then differentiating with respect to time, I obtain the evolution of relative prices in sector k as a function of sectoral TFP growth rates and changes in the relative unit cost of input in sector j,

$$\frac{\dot{p_k(t)}}{p_k(t)} = \frac{\dot{A_H(t)}}{A_H(t)} - \frac{\dot{A_k(t)}}{A_k(t)} + \frac{\dot{\Gamma_k(t)}}{\Gamma_k(t)}. \tag{B.4.6}$$

Since the TFP growth rates are constant over time (i.e. $g_j(t) = g_j$), this means

$$\frac{\dot{p_k(t)}}{p_k(t)} = g_H - g_k + \frac{\dot{\Gamma_k(t)}}{\Gamma_k(t)}. \tag{B.4.7}$$

This implies that two forces shape the evolution of relative prices over time. The first is the relative TFP growth rates; a higher TFP growth rate in sector k reduces the relative price of that sector. The second is the evolution of the relative unit cost of input in sector k; a faster growth rate in the unit cost of input in sector H reduces the relative price of sector k over time. From equation (B.4.5), it is clear that if the wage rate of skilled labor, $w_S(t)$, is rising, this will raise the unit cost of input in the skill-intensive sector. Equations (B.4.3) and (B.4.7) allow me to proceed to the main proof.

Main proof. To proceed, I replace the expression that pins down the evolution of relative prices, (B.4.7), inside the expression that pins down the evolution of relative expenditure shares, (B.4.3). This means that the relative expenditure share of sector k evolve over time according to

$$\frac{\dot{R_k(t)}}{R_k(t)} = (1-\sigma)\left(g_H - g_k\right) + (1-\sigma)\left(\lambda_k - \lambda_H\right)\frac{\dot{c_i^g(t)}}{c_i^g(t)} + (1-\sigma)\frac{\dot{\Gamma_k(t)}}{\Gamma_k(t)}. \tag{B.4.8}$$

Expression (B.4.8) reveals that relative expenditure shares evolve according to three factors whose sign depends on whether sectoral preferences are gross substitutes ($\sigma > 1$) or complements ($\sigma < 1$). The proposition assumes the latter. The first depends on the relative TFP growth rates; faster TFP growth in sector k relative to sector H reduces the expenditure share of sector k relative to sector H as per the standard Baumol cost disease mechanism.

The second depends on relative income elasticity: if sector k is the income-inelastic sector relative to sector H (i.e. $\lambda_k < \lambda_H$), then as households get richer they bias their consumption away from sector k. The first two are given thanks to the assumptions of the proposition that $\sigma < 1$, $g_H < g_L, g_G$ and $\lambda_H > \lambda_L, \lambda_G$. The third is the relative input cost of sector k: if the input cost of producing a single unit of sector k output is falling relative to that of sector H (either due to differences in skill intensity, the evolution of the wage rate premium or both), then this would reinforce the decline in the relative price of sector k, thereby reducing its expenditure share. Given that the first two forces are a function of parameters, they are bounded at x_k such that

$$-x_k \geq (1-\sigma)\left(g_H - g_k\right) + (\lambda_k - \lambda_H)(1-\sigma)\frac{\dot{c_i^g}(t)}{c_i^g(t)}. \tag{B.4.9}$$

Let $\Delta_k(t) \equiv (1-\sigma)\frac{\dot{\Gamma_k}(t)}{\Gamma_k(t)}$ and $f_k(t) \equiv \log(R_k(t))$ such that $\dot{f}_k(t) = \frac{\dot{R_k}(t)}{R_k(t)} \quad \forall k \in \{L, G\}$. The fundamental theorem of calculus states that

$$f_k(t) - f_k(0) = \int_0^t \dot{f}_k(s)ds. \tag{B.4.10}$$

Replacing (B.4.8) and making use of the definitions of $f_k(t)$ and $\Delta_k(t)$ into (B.4.10) yields

$$f_k(t) - f_k(0) = \int_0^t \left((1-\sigma)\left(g_H - g_k\right) + (1-\sigma)(\lambda_k - \lambda_H)\frac{\dot{c_i^g}(s)}{c_i^g(s)}\right)ds + \int_0^t \Delta_k(s)ds \tag{B.4.11}$$

The first integrand is bounded at every s to $-x_k$ as per (B.4.9) implying that it can be reduced to a bound of $-x_k t$ at each single time period t. The second integral, also according to the fundamental theorem of calculus, is

$$\int_0^t \Delta_k(s)ds = (1-\sigma)\left(log\Gamma_k(t) - log\Gamma_k(t)\right). \tag{B.4.12}$$

The term $\Gamma_k(t)$ is bounded. This is because

$$\frac{\phi_k^\epsilon + (1-\phi_k)^\epsilon\, y(t)}{\phi_H^\epsilon + (1-\phi_H)^\epsilon\, y(t)} \quad \text{where} \quad y(t) \equiv \left(\frac{w_S(t)}{B(t)\, w_N(t)}\right)^{1-\epsilon} > 0, \tag{B.4.13}$$

which is simply a weighted average of $(\phi_k/\phi_H)^\epsilon$ and $\left((1-\phi_k)/(1-\phi_H)\right)^\epsilon$. Both are positive and finite since $0 < \phi_j < 1$. Hence $\underline{\Gamma}_k \leq \Gamma_k(t) \leq \overline{\Gamma}_k$ for all t, and

$$f_k(t) - f_k(0) \leq -x_k t + (1-\sigma)\log\frac{\overline{\Gamma}_k}{\underline{\Gamma}_k} \longrightarrow -\infty \quad \text{as} \quad t \longrightarrow \infty. \tag{B.4.14}$$

C. Quantitative Analysis

C.1 Numerical Solution

To solve the model numerically, I adapt the algorithm in Buera and Shin (2017). The algorithm is composed of two parts. The first part computes the steady state value function and distribution based on a stationarized, one-sector version of the where a single sector dominates (as per Section 3.4). The second part computes the transition by iterating over wage sequence until convergence using the steady state as the terminal condition.

The state space is 2-dimensional, defined on parent-child human capital pairs. In recognition that the support for initial childhood human capital grids may differ from the parent human capital support, I distinguish between $\mathcal{H}$ and $\mathcal{H}_0'$ where the former is the support for parent human capital and the latter is the support for child human capital obtained from the initial draws. The state space is therefore defined along the $\mathcal{H} \times \mathcal{H}_0'$ dimensions.

To proceed, I discretize the parent human capital support $\mathcal{H}_d = \{h_1, ..., h_K\}$ and the child human capital draw support $\mathcal{H}_{0,d}' = \left\{h_{0,1}', ..., h_{0,K}'\right\}$. I also discretize the initial child human capital draw probability density function $g(\eta_c, \kappa_c)$ into a probability mass function $g_d(\eta_c, \kappa_c)$. Finally, I generate sequences of $A_j(t)$ which are constructed by feeding the constant sectoral TFP growth rates g_j into a compound growth function.

Steady State. Algorithm 1 presents the solution method for solving for the steady state in a single-sector economy with only High-Skill Services sector output being consumed and produced. This is done in three steps. The first step performs value function iteration over the generation's problem for each parent human capital-child draw pairs. This step pins down the optimal policy function for each pair and the terminal condition, which is the value function in steady state V^{ss}. The second step iterates over the parent human capital distribution using the optimal time allocation policy function and the initial child human capital draws. The third step ensures that the market for High-Skill Services sector output clears. Both consumption and skills supply are aggregated using the parent human capital and initial child human capital distributions as weights.

Algorithm 1 Steady State

Require: Grids $\mathcal{H}_d, \mathcal{H}'_{0,d}$, child pmf $g_d(\eta_c, \kappa_c)$, choice set $[\underline{\tau}, \bar{\tau}]^S$, discount factors β, β_g, measure of households $\mathcal{L}$, tolerances ε, ε_μ, and maximum iterations $i_{\max}$.

Ensure: V^{ss}, μ^{ss}, and $\boldsymbol{\tau}^{*ss}$.

1: Stationarize wages: $w(t)/A(t) = 1$ for all $t > \mathcal{T}$.

2: Initialize $V^{(1)}$, set $i \leftarrow 1$ and $\Delta_V \leftarrow \infty$.

3: **while** $\Delta_V > \varepsilon$ and $i < i_{\max}$ **do**

4: **for** $(h, h'_0) \in \mathcal{H}_d \times \mathcal{H}'_{0,d}$ **do**

5: Solve, by backward induction over $s = S, \ldots, 1$,

$$\widetilde{V}(h, h'_0) = \max_{\boldsymbol{\tau} \in [\underline{\tau}, \bar{\tau}]^S} \left\{ \sum_{s=1}^{S} \beta^{s-1} u(c_s, \tau_s) + \beta^S \beta_g \, \mathcal{I}[V^{(i)}]\big(h'(h, h'_0, \boldsymbol{\tau})\big) \right\}, \quad c_s = (1 - \tau_s)h,$$

and store $\boldsymbol{\tau}^{*ss}(h, h'_0)$ and $\boldsymbol{c}^{*ss}(h, h'_0)$.

6: **end for**

7: $V^{(i+1)}(h) \leftarrow \sum_{h'_0 \in \mathcal{H}'_{0,d}} g_d(h'_0) \widetilde{V}(h, h'_0)$.

8: $\Delta_V \leftarrow \max_{h \in \mathcal{H}_d} |V^{(i+1)}(h) - V^{(i)}(h)|$; $i \leftarrow i + 1$.

9: **end while**

10: $V^{ss} \leftarrow V^{(i)}$.

11: Construct Π^{*ss} from $\boldsymbol{\tau}^{*ss}$, set $j \leftarrow 1$ and iterate $\mu^{(j+1)} = \Pi^{*ss} \mu^{(j)}$ until $\max_{h \in \mathcal{H}_d} |\mu^{(j+1)}(h) - \mu^{(j)}(h)| < \varepsilon_\mu$; set $\mu^{ss} \leftarrow \mu^{(j)}$.

12: **for** $s = 1, \ldots, S$ **do**

13: $c_s^{ss} \leftarrow \mathcal{L} \sum_{h, h'_0} \mu^{ss}(h) g_d(h'_0) c_s^{*ss}(h, h'_0)$.

14: $Y_s^{ss} \leftarrow \mathcal{L} \sum_{h, h'_0} \mu^{ss}(h) g_d(h'_0) h \big(1 - \tau_s^{*ss}(h, h'_0)\big)$.

15: Check $|c_s^{ss} - Y_s^{ss}| < \varepsilon$.

16: **end for**

Transition. Algorithm 2 presents the solution method applied to solve for the transition. Crucially, this is achieved by iterating over the fixed point sequence of the wage rate of unskilled labor, $\{w_N(t)\}_{t=0}^{\mathcal{T}}$, until the market for unskilled labor clears. Conditional on a path for the wage rate of unskilled labor, the paths of the skilled labor wage rate, the two sectoral prices and the occupational choice thresholds are identified.

Algorithm 3 is called in line 6. This algorithm takes as inputs the proposed path of wages, prices and the occupational choice threshold, solves the household problem and computes the joint distribution and, accordingly, the aggregate quantities $c_j(t), S(t), N(t)$ $\forall t < \mathcal{T}$, $j \in \{G, H, L\}$. This is done in three steps. The first step takes as input the steady state value

function solved in Algorithm 1 and, through backward induction, computes optimal policy for each household human capital pairs. The second step uses the optimal time allocation policy and shoots forward using the discrete-time Kolmogorov Forward Equation (KFE) to solve for the joint distribution in all generations. The third step uses the joint distribution to compute equilibrium aggregates including the supply of skills in the economy and sectoral output demand given the proposed price paths.

Algorithm 2 Transition-Path Equilibrium: Price Block

Require: TFP paths $\{A_j(t)\}_{t\in\mathcal{D}}$, $j \in \{G, H, L\}$, SBTC path $\{B(t)\}_{t\in\mathcal{D}}$, terminal value V^{ss}, initial distribution μ_1, maximum step $\omega_{\max}$, tolerance ε_N, and maximum iterations $i_{\max}$.

Ensure: Equilibrium paths $\{w_N(t), w_S(t), p_G(t), p_L(t)\}_{t\in\mathcal{D}}$.

Set $p_H(t) \leftarrow 1$ for all $t \in \mathcal{D}$. ▷ Numeraire

Initialize $w_N^{(1)}$, $\omega \leftarrow \omega_{\max}$, $i \leftarrow 1$, and $\Delta \leftarrow \infty$.

while $\Delta > \varepsilon_N$ and $i \leq i_{\max}$ **do**

  Given $w_N^{(i)}(t)$, obtain $w_S(t)$ from zero profit in the H-sector, and $p_G(t)$, $p_L(t)$ from unit costs.

  Set $\bar{h}(t) \leftarrow w_N^{(i)}(t)/w_S(t)$.

  Obtain $c_j(t)$, $S(t)$, and $N(t)$ from Algorithm 3.

  Back out sectoral labour demands $N_j(t)$, $S_j(t)$ from $Y_j(t) = c_j(t)$, and set $N^d(t) \leftarrow \sum_j N_j(t)$.

  $e(t) \leftarrow N^d(t)/N(t) - 1$ and $\Delta \leftarrow \max_{t\in\mathcal{D}} |e(t)|$.

  If Δ rose, set $\omega \leftarrow \max\{0.02, \omega/2\}$; otherwise $\omega \leftarrow \min\{\omega_{\max}, 1.25\,\omega\}$.

  $w_N^{(i+1)}(t) \leftarrow w_N^{(i)}(t) \left[N^d(t)/N(t)\right]^{\omega}$, $t \in \mathcal{D}$.

  $i \leftarrow i + 1$.

end while

The skilled labour market clears by Walras' law.

Algorithm 3 Transition-Path Equilibrium: Household and Distribution Block

Require: Candidate paths $\{w_N(t), w_S(t), p_G(t), p_L(t), \bar{h}(t)\}_{t\in\mathcal{D}}$, terminal value V^{ss}, initial distribution μ_1, grids $\mathcal{H}_d, \mathcal{H}'_{0,d}$, child pmf $g_d(\eta_c, \kappa_c)$, choice set $[\underline{\tau}, \bar{\tau}]^S$, discount factors β, β_g, and measure of households $\mathcal{L}$.

Ensure: $\{c_j(t), S(t), N(t)\}_{t\in\mathcal{D}}$, $j \in \{G, H, L\}$.

Set $\overline{V}_G(h) \leftarrow V^{\text{ss}}(h)$.
for $g = G-1$ **down to** 1 **do**
  for $h \in \mathcal{H}_d$ **do**
    for $s = 1, \ldots, S$ **do**
      Set $t \leftarrow (g-1)S + s$.
      Determine $o^*(h;t)$ and gross income $w_s(h;t)$.
    end for
    for $h'_0 \in \mathcal{H}'_{0,d}$ **do**
      Solve, by backward induction over $s = S, \ldots, 1$,

$$V_g(h, h'_0) = \max_{\boldsymbol{\tau}\in[\underline{\tau},\bar{\tau}]^S} \left\{ \sum_{s=1}^{S} \beta^{s-1} u(c_s, \tau_s) + \beta^S \beta_g \, \mathcal{I}[\overline{V}_{g+1}]\big(h'(h, h'_0, \boldsymbol{\tau})\big) \right\},$$

  where c_s solves $p(c_s;t)\, c_s = (1-\tau_s) w_s(h;t)$.
      Store $\boldsymbol{\tau}^*(h, h'_0; g)$, $\boldsymbol{c}^*(h, h'_0; g)$, and $c^*_{j,s}(h, h'_0; g)$ from the sectoral demands.
    end for
    $\overline{V}_g(h) \leftarrow \sum_{h'_0 \in \mathcal{H}'_{0,d}} g_d(h'_0) V_g(h, h'_0)$.
  end for
end for
Initialize $\mu_1(h, h'_0) \leftarrow \mu_1(h) g_d(h'_0)$.
for $g = 1, \ldots, G-2$ **do**
  Propagate μ_g through $h'(h, h'_0, \boldsymbol{\tau}^*(h, h'_0; g))$ to obtain $\mu_{g+1}(h)$.
  Set $\mu_{g+1}(h, h'_0) \leftarrow \mu_{g+1}(h) g_d(h'_0)$.
end for
for $t \in \mathcal{D}$ **do**
  Set $g \leftarrow \lceil t/S \rceil$ and $s \leftarrow t - (g-1)S$.
  $S(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) h (1 - \tau^*_s) \mathbf{1}\{h > \bar{h}(t)\}$.
  $N(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) (1 - \tau^*_s) \mathbf{1}\{h \le \bar{h}(t)\}$.
  $c_j(t) \leftarrow \mathcal{L} \sum_{h, h'_0} \mu_g(h, h'_0) c^*_{j,s}(h, h'_0; g)$, $j \in \{G, H, L\}$.
end for

C.2 The Timing Convention

Table C.1 connects generations and stages of childhood development that are used to build the synthetic cohorts for the time use data with calendar years. That way, the generation-level moments such as skilled worker and expenditure shares are 15-year averages over the same calendar years that define a generation. This ensures consistency with the stage-specific childcare time moments which partition that window into three 5-year stages of childhood development.

Table C.1. Linking Generations, Stages of Childhood Development, and Calendar Years

Generation	Stage of development	Calendar years	Age of child (years, inclusive)
1	1	1942–1946	0–4
	2	1947–1951	5–9
	3	1952–1956	10–14
2	1	1957–1961	0–4
	2	1962–1966	5–9
	3	1967–1971	10–14
3	1	1972–1976	0–4
	2	1977–1981	5–9
	3	1982–1986	10–14
4	1	1987–1991	0–4
	2	1992–1996	5–9
	3	1997–2001	10–14
5	1	2002–2006	0–4
	2	2007–2011	5–9
	3	2012–2016	10–14
6	1	2017–2021	0–4
	2	2022–2026	5–9
	3	2027–2031	10–14

Notes: This table maps each model generation and stage of childhood development into the corresponding calendar years and age range. Each generation contains three five-year stages of childhood development: stage 1 covers ages 0–4, stage 2 covers ages 5–9, and stage 3 covers ages 10–14. The mapping is used to identify generation- and stage-specific empirical moments and to construct the synthetic cohorts underlying the parental time-allocation moments.

C.3 Constructing Targeted Data Moments

Time Allocation. I obtain the time allocation moments from the Multinational Time Use Survey (Lamote de Grignon Pérez et al., 2024), restricting attention to the US samples which are available from 1965 onward. I keep individuals who are (i) in full-time employment, (ii) between the ages of 21 and 64 (inclusive) and (iii) live in a household that contains a child below the age of 15.[26] To remain consistent with the model, the parental childcare time shares are defined as the time allocated to childcare divided by the time allocated to both childcare and paid work.

Since there exists a single cohort of both parents and children at any given time in the model, all children in the economy are in the same stage of childhood development. I therefore proceed by reconstructing time allocation profiles for synthetic cohorts using age-restricted cross sections. Tracking the time allocation of parents of a given cohort over time (using the timing convention Appendix C.2) enables identification of stage-specific elasticities of parental time allocation. Where multiple calendar years are available for a given parent generation-child stage of development period, I take a simple average. Diarist-day observations for each sample are weighted using the proposed survey weights. The three time allocation moments are:

Generation 3, stage 1 (1972-1976)

- Time allocated to childcare for diarists who live in a household with a child between 0-4 years old

Generation 6, stage 1 (2017-2021)

- Time allocated to childcare for diarists who live in a household with a child between 0-4 years old

Generation 3, stage 3 (1982-1986)

- Time allocated to childcare for diarists who live in a household with a child between 10-14 years old

Skill Distribution. The skill distribution data moments are computed using microdata obtained from IPUMS USA (Ruggles et al., 2025). Attention is restricted to individuals who are (i) employed, (ii) between the ages of 21 and 64 (inclusive), (iii) work at least

[26] The surveys do not capture how the diarist relates to the child.

30 hours a week and (iv) worked for at least 40 weeks.[27] Skilled workers are defined as workers whose occupation is either managerial, professional or technical (Acemoglu and Autor, 2011), where the occupation identifier variable is OCC1950. Skills are assumed fixed after childhood, meaning that the skill distribution is a generation-specific distribution. Thus, where multiple sample calendar years are available for a single generation, I take a simple average. All skill distribution moments are weighted using person weights. The four targeted skill distribution moments are:

Generation 1, 1942-1956

- Share of skilled workers
- Log 90-10 wage per hour ratio of skilled workers

Generation 6, 2017-present

- Share of skilled workers
- Log 90-10 wage per hour ratio of skilled workers

Structural Change. The structural change moments are sectoral expenditure shares, the Goods sector relative price and the sector-specific wage bill share paid to skilled workers. Starting with the sectoral expenditure shares, the production function does not feature intermediate inputs. Thus, to consistently map consumption expenditures to the data, it is necessary to convert final expenditure by sector into consumption value added by sector. Herrendorf, Rogerson and Valentinyi (2013) lay out the procedure to do this in their Online Appendix. For the sake of completeness, I provide a brief summary of these steps in Appendix C.4 together with the time series of consumption value added for my three broad sectors of interest. The procedure requires input-output tables and the total requirements matrix which are made available by the BEA for benchmark years (U.S. Bureau of Economic Analysis, 2025). Where data availability permits, I average over the 15-window for a given generation in order to remove stage-specific trends and isolate cross-sectional variation in sectoral expenditure shares.

Aside from sectoral expenditure shares, I also target the change in the Goods relative price. This requires data on sectoral value added prices, which I construct from the

[27] Hours worked per week are usual hours worked per week (i.e. UHRSWORK) where available, hours worked last week (i.e. HRSWORK1) otherwise, and the midpoint of the intervalled variable HRSWORK2 for the remaining samples. Weeks worked are WKSWORK1 where available and the midpoint of the intervalled variable WKSWORK2 otherwise. Hours worked per year are the product of hours worked per week and weeks worked.

industry-level production accounts of Elbadri and Vizcaino (2025). For each industry, the value added price is nominal value added (normalized to unity in 1947) divided by its value added quantity index (normalized to unity in the same year). Sectoral prices are Törnqvist aggregates of the industry prices within each sector, and the relative price of Goods is the ratio of the Goods value added price to the High-Skill Services value added price.

Finally, I also target the sectoral wage bill shares paid to skilled workers. This is built from the same dataset that is used to obtain moments on the skill distribution Ruggles et al. (2025). Skilled workers are again proxied by occupation and the skilled workers' wage bill share in a given sector is the sum of annual wage and salary earnings of skilled workers divided by the sum of annual wage and salary earnings of all workers. Combined, these result in a total of eight targeted structural change moments:

Generation 1, 1942-1956

- High-Skill Services expenditure share
- Goods expenditure share
- High-Skill Services skilled workers' wage bill share
- Low-Skill Services skilled workers' wage bill share
- Goods skilled workers' wage bill share

Generation 6, 2017-present

- High-Skill Services expenditure share
- Goods expenditure share

Generation 6, stage 1 (2017-2021)

- Change in the Goods value added price (relative to High-Skill Services value added price) compared with generation 2, stage 1 (1957-1961)

C.4 Computing Consumption Value Added

Herrendorf, Rogerson and Valentinyi (2013) offer a complete description of the steps required to extract sectoral consumption value added from input-output tables in the Online Appendix B. Given that I distinguish between High- and Low-Skill Services and aggregate Agriculture and Manufacturing into Goods, I repeat this exercise using the same input-output tables but with the relevant sectoral split. I provide a brief description of the steps below for completeness.

Let m be the number of commodities produced by the economy and n be the number of industries that produce these m commodities. In input-output tables prior to 1972, $m = n$.

1. From the Use table, extract the final consumption expenditure vector $\mathbf{e}_c$. This is an $m \times 1$ vector where each element i denotes the value of final consumption expenditure spent on the corresponding commodity.

2. From the Use table, extract total output and total value added by industry. Compute the value added share of output by taking the ratio of total value added over total output for each industry. This vector $\mathbf{v}$ is an $n \times 1$ vector.

3. Extract the total requirements matrix $\mathbf{R}$. In post-1972 tables, this is an $n \times m$ matrix. Entry ji of the total requirements matrix gives the dollar value of industry j's production that is required to deliver one dollar of the domestically produced commodity i to final uses.

4. Consumption value added by industry is then

$$\mathbf{c} = \langle \mathbf{v} \rangle \mathbf{R}\, \mathbf{e}_c$$

   where $\langle \mathbf{v} \rangle$ is the diagonal matrix with vector $\mathbf{v}$ on its diagonal. Element i of $\mathbf{c}$ denotes the consumption value added of the corresponding industry.

5. Aggregate industries into sectors by summing the elements of $\mathbf{c}$ corresponding to each sector as per the industry classification in Section 2. Repeat these steps for each year where input-output tables are available.

These steps are performed using the BEA's Historical Benchmark Input-Output Tables for years where data is available (1947, 1958, 1963, 1972, 1977, 1987, 1992) as well as the annual Input-Output Accounts Data releases for later years (1997-2024).

C.5 Moment Sensitivity Validation

Table C.2 reports, for each parameter, the targeted moment most sensitive to it at the SMM estimate. Every parameter moves at least one moment with an elasticity of 0.46 or more in absolute value, and the Jacobian of the moments with respect to the parameters has full rank.

Table C.2. Identification of the Internally Calibrated Parameters

Parameter	Description	Value	Most sensitive moment	Elasticity
Panel A. Structural Change				
θ_G	Consumption weight, G-sector	0.433	G-sector expenditure share (Gen. 6)	1.09
θ_H	Consumption weight, H-sector	0.254	H-sector expenditure share (Gen. 1)	1.13
λ_G	Income elasticity parameter, G-sector	0.970	Childcare time, stage 3 (Gen. 3)	0.46
λ_H	Income elasticity parameter, H-sector	2.128	H-sector expenditure share (Gen. 1)	−0.73
ϕ_H	Unskilled labor intensity, H-sector	0.358	Skilled share of H-sector wage bill (Gen. 1)	−1.69
ϕ_L	Unskilled labor intensity, L-sector	0.472	Skilled share of L-sector wage bill (Gen. 1)	−3.39
ϕ_G	Unskilled labor intensity, G-sector	0.508	Skilled share of G-sector wage bill (Gen. 1)	−4.06
g_B	Growth rate of skill-biased technical change	0.103	Skilled worker share (Gen. 6)	0.60
Panel B. Time Use				
ξ_0	Level, human capital technology	−1.367	Childcare time, stage 3 (Gen. 3)	−7.68
ξ_1	Stage gradient, human capital technology	0.675	Childcare time, stage 3 (Gen. 3)	−7.17
ν	CRRA parameter	1.943	Childcare time, stage 1 (Gen. 3)	−0.85
Panel C. Skill Supply				
η	Location, initial parental distribution	−2.298	Skilled share of G-sector wage bill (Gen. 1)	1.63
κ	Shape, initial parental distribution	1.414	Skilled share of G-sector wage bill (Gen. 1)	1.44
η_c	Location, child endowment distribution	−3.445	Childcare time, stage 1 (Gen. 3)	1.66
κ_c	Shape, child endowment distribution	1.119	Childcare time, stage 3 (Gen. 3)	0.53

Notes: Elasticity is $\partial \log m / \partial \log |\theta|$ of the most responsive targeted moment at the estimate, from ±5% central differences (one-sided near a bound). Gen. g denotes generation g; H, L and G denote High-Skill Services, Low-Skill Services and Goods.